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Data augmentation: A comprehensive survey of modern approaches

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ABSTRACT

To ensure good performance, modern machine learning models typically require large amounts of quality annotated data. Meanwhile, the data collection and annotation processes are usually performed manually, and consume a lot of time and resources. The quality and representativeness of curated data for a given task is usually dictated by the natural availability of clean data in the particular domain as well as the level of expertise of developers involved. In many real-world application settings it is often not feasible to obtain sufficient training data. Currently, data augmentation is the most effective way of alleviating this problem. The main goal of data augmentation is to increase the volume, quality and diversity of training data. This paper presents an extensive and thorough review of data augmentation methods applicable in computer vision domains. The focus is on more recent and advanced data augmentation techniques. The surveyed methods include deeply learned augmentation strategies as well as feature-level and meta-learning-based data augmentation techniques. Data synthesis approaches based on realistic 3D graphics modeling, neural rendering, and generative adversarial networks are also covered. Different from previous surveys, we cover a more extensive array of modern techniques and applications. We also compare the performance of several state-of-the-art augmentation methods and present a rigorous discussion of the effectiveness of various techniques in different scenarios of use based on performance results on different datasets and tasks.

1. Introduction

1.1. Background

Machine learning algorithms enable computing systems to learn to solve various problems based on observed data. Despite enormous progress in deep learning algorithms in recent years, obtaining adequate and representative training data remains the most important factor that determines the performance of machine learning models. Irrespective of innovations in model design and training techniques, training with small and poorly representative data will ultimately lead to poor generalization performance. On the other hand, rich, large and more representative data invariably guarantees good performance, even when using less sophisticated algorithms [1]. For many practical tasks, obtaining sufficiently large and representative data remains a challenging task. Several studies [2–4] have shown that for many tasks even today's large-scale datasets such as ImageNet [5] and CIFAR-100 [6] lack adequate level of variability to guarantee good performance. Increasing the robustness and accuracy of machine learning models and allowing them to perform well on small, poorly representative data is the main goal of data augmentation. Recent works show that these shortcomings can be mitigated by exploiting appropriate data augmentation techniques, where additional training samples are generated from existing training data or created from scratch using various techniques. Effective data augmentation strategies can also reduce the requirements on algorithmic sophistication and enable simpler model architectures to achieve high generalization performance [1]. In the case of computer vision, training data is typically in the form of images or video sequences. In most tasks, a video sequence is treated as a frame-by-frame collection of images. Three-dimensional (3D) computer vision tasks typically rely on voxel and point cloud data.

The larger, more diverse and representative (with respect to the distribution of the target dataset) the training data, the more effective the deep learning model performs on unobserved data. On one hand, the diversity of training samples should be sufficient for the model to handle different deviations in image appearance and even noisy target instances. On the other hand, it is necessary that the quality does not degrade so much as to hamper performance on normal images.

1.2. Status of data augmentation and related concepts

Data augmentation is part of a broad set of regularization techniques aimed at improving model performance. Regularization methods work

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by introducing additional information to the underlying machine learning model to better capture more general properties of the problem being modeled. The simplest and one of the most widely used approaches is to introduce a penalty term to the cost function to constrain the values of model parameters. This allows the trained model to have a more stable and smoother response to various variations of input data (at test time). Batch normalization [7] normalizes the outputs of various layers so as to eliminate the effects of internal covariance shift in inputs to deep layers. Regularization can also be achieved by varying the effective size and structure of the deep learning model (e.g., model pruning [8,9]), changing the number of learnable parameters [10], or the pattern of synaptic connections (e.g., dropconnect [11]). Dropout regularization [12,13], a strategy closely related to dropconnect, relies on eliminating some network units during training. This family of methods are aimed at reducing model complexity so that unnecessary patterns are not learned from the data. Another common class of techniques involve changing the training procedure (e.g., early stopping [14]). Regularization methods such as weight decay [15] provide an easy way to control overfitting of large neural network models.

An alternative workaround in situations where large training data is difficult to access is by the use of transfer learning (TF) and fine-tuning [16,17]. The approach re-uses a pre-trained model that has been trained on a sufficiently large dataset on a new but related task, thus enabling the model to generalize to the new context without extensive training from scratch. Transfer learning techniques have been successfully used with small datasets to achieve good generalization performance.

Another common approach (e.g., in [18–20]) is to focus on developing models that rely on more effective feature representations of the training data to increase performance. An important class of methods in this line of works is representation learning [21,22].

Data augmentation, the subject of the current work, relates to methods that seek to improve performance by changing some characteristics of training dataset itself.

1.3. General problem of data augmentation

For a given dataset D consisting of training samples S_i with corresponding labels L_i, the task of data augmentation is to compose transformation operations T_j that can be applied to the original samples S_i to create additional training data S_i' without altering the corresponding labels. These types of warping operations are known as label preserving transformation operations in image processing and computer vision literature [23]. That is, the transformations $T_j\left(S_i\right)$ results in a modified sample that can still be semantically described by the original label L_i. It is worth noting that there is, however, currently no objective measure of "labelness" and, hence, what constitutes label preservation. The standard practice is to leverage human knowledge for the assessment. An image label is understood to be preserved under transformation if its original meaning is still recognizable by a human agent. The general idea is that for training samples belonging to a given n-dimensional sample space $S_i \in R^n$, the transformed samples S_i' also belong to the same sample space as the original data S_i, i.e., $T_j : R^n \rightarrow R^n$, and the transformed samples are assigned the original labels L_i: $T\left(S_i, L_i\right) \rightarrow \left(S_i', L_i\right)$

Multiple transformation operations can be performed to enhance diversity. Owing to limitations on computational resources, cost and undesirable effects of spurious data, it is generally advisable to select only the most useful augmentations.

1.4. Similar surveys

Despite the importance of data augmentation in computer vision, the number of extensive surveys on data augmentation methods has been quite limited. Indeed, only in the last four years has there been notable surveys on data augmentation techniques for computer vision

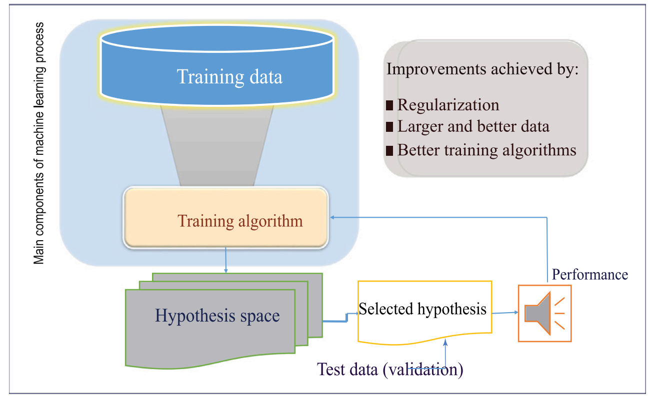

Fig. 1. Main components and workflow of the machine learning process.

applications. Although not specifically a survey paper, Mikołajczyk and Grochowski [24] in 2018 presented an elaborate discussion of data augmentation approaches to improving performance in image classification tasks. Shorten and Khoshgoftaar [25] in 2019 introduced one of the earliest and most extensive reviews of data augmentation techniques in computer vision. It is still arguably one of most comprehensive surveys till date. In the work [25], the authors presented a detailed treatment of a wide range of data augmentation methods for 2D image data. Their review covers many pertinent issues that arise in the process of applying data augmentation. These include changes in image resolution and explosion of dataset size. However, the work does not provide an in-depth coverage of more recent data augmentation methods such as neural style transfer (NST), 3D modeling, differential neural rendering, generative modeling and meta-learning approaches. It is also worth noting that the above survey, despite its broad coverage, is primarily focused on image classification tasks. Khalifa et al. [26] provide a more concise presentation of common 2D image data augmentation methods covering a more extensive range of computer vision domains. The use of data augmentation in different application domains – specifically, medical imaging, remote sensing and agriculture – is particularly highlighted. However, recent state-of-the-art methods such as neural style transfer, meta-learning and GAN-based approaches are only briefly covered.

A number of less extensive surveys have also been conducted on data augmentation in general computer vision settings. Khosla and Saini's study [27], apart from providing a brief coverage of GAN techniques, has mainly tackled traditional image manipulation-based data augmentation techniques. Alternative workarounds aimed at improving generalization performance of computer vision models that do not involve data augmentation are also concisely introduced. The study in [28] is largely focused on a subset of data augmentation approaches — what we termed in this survey as region-level data augmentation methods: image region deletion, addition and combination approaches. Yang et al. in [29] present a brief survey of image data augmentation methods as well as comparative results of common data augmentation techniques in image classification, object detection and segmentation tasks. Kaur et al. [30] mainly focus on data augmentation in object detection tasks. In contrast to the above works, we provide a thorough coverage of many common data augmentation approaches, including both traditional and recently introduced state-of-the-art methods not covered in these works.

While a large number of the works (e.g., [25–27,29]) deal with data augmentation in general computer vision tasks, many recent studies have focused on narrower application domains such as medical image analysis [31–34], face recognition [35], pose estimation [36], sentiment analysis [37], text recognition [38,39] and remote sensing [40,41]. This treatment is justified on the basis of the fact that machine learning has become highly specialized and more fragmented, resulting in specific techniques and datasets being introduced to solve specific problems. However, such works generally tend to focus on domain-specific techniques while de-emphasizing more useful general concepts with broad applications. On the other hand, some surveys such as [42] have also tackled the problem of data augmentation as a generic data processing and regularization step required for correcting deficiencies in training data. These works generally address common techniques that are applicable to a wide range of machine learning tasks. As a result of their generic approach, peculiar issues relevant to different machine learning domains are usually not treated. Moreover, their approach present difficulties in selecting the most suitable methods when dealing with specific tasks. One of the main aims of our work is to bridge this gap by providing useful insights regarding the most important techniques that work well in general computer vision tasks while highlighting the important considerations and specific strengths and limitations in different scenarios.

1.5. Main contributions

In this paper, we provide an extensive and thorough coverage of recent data augmentation methods in a wide range of computer vision domains, significantly extending the scope of previous surveys in terms of both breadth and depth. Based on a new taxonomy of approaches, we present a novel perspective on data augmentation methods that reflect current practice, and provide many relevant examples of works in line with the ideas and concepts discussed. We explore the peculiar issues of the application of data augmentation in common computer vision tasks. We emphasize specific use-cases of the different classes of approaches, highlighting their particular strengths and weaknesses to guide researchers and machine learning practitioners on the selection of suitable data augmentation methods for different applications. Importantly, in this regard, we include extensive quantitative performance comparison of various data augmentation methods. In summary, the main contributions of this study are that:

- we introduce a new taxonomy of approaches that reflects current state of data augmentation research and practice. Specifically, we divide all data augmentation methods into two broad classes of approaches — data transformation and data synthesis. The emergence of several new augmentation approaches based on data synthesis techniques makes such a categorization rational. Each of these broad family of methods is further subdivided according to the common techniques, application scenarios and emerging trends. Furthermore, we treat meta-learning approaches as distinct class of methods that can be utilized to achieve data transformation and data synthesis-based augmentation.
- we provide an in-depth and extensive, up-to-date presentation of data augmentation methods relevant to computer vision tasks, including approaches that utilize non-standard image modalities such as thermal and infrared images, as well as methods applicable in 3D computer vision settings. We particularly focus on more modern data augmentation methods.
- we present extensive quantitative performance results of several data augmentation methods on benchmark datasets. The results are summarized under three broad categories – generic data augmentation techniques, advanced data augmentation methods and combined augmentations – composite methods that leverage two or more augmentation techniques simultaneously. To the best of our knowledge, this paper is the first work to present such an extensive comparison of the predictive performance of data augmentation methods.
- based on the thorough review of recent state-of-the-art data augmentation methods, we outline major challenges and promising directions for future research.

1.6. Taxonomy of data augmentation approaches and outline of survey

In this paper, we adopt a taxonomy of data augmentation methods quite distinct from all known previous surveys. We contend that the main way to extend training data is either by manipulating existing training data or by creating new data samples "from scratch". Based on this viewpoint, we divide all data augmentation approaches into two: (1) so-called data transformation approaches and (2) data synthesis methods. Data transformation approaches cover all classical data augmentation methods, as well as more recent approaches that work by warping input images or feature maps to produce variability. Data synthesis methods basically create new training samples "from scratch".

We also distinguish meta-learning as a way to either extend the generality of data representation or to automate the process of learning useful data transformations in order to expand the sample space. In either case, meta-learning, as a data augmentation approach, can be viewed as a distinct category that has a direct relationship with the two broad categories in the sense that it can be directly applied to enhance the generality of these methods or to automate the data augmentatin process. This special relationship is shown in Fig. 2.

Further, we classify data transformation methods into two broad categories: input space and feature space methods. Input space approaches include traditional image transformation (i.e., photometric and geometric) methods as well as more recent techniques based on learned image manipulation strategies. We categorize the latter methods as advanced input space data augmentation techniques. Feature space methods rely on manipulating deep convolutional neural network (CNN) feature maps to generate new, transformed data for training.

The second broad class of approaches are data synthesis techniques. These include computer graphics modeling, neural rendering, neural style transfer, as well as generative modeling techniques. Fig. 2 illustrates the detailed taxonomy of approaches used in this survey.

2. Input space data augmentation

Input space (or sample space) data augmentation techniques are a class of approaches that directly modify the input image (or parts of it) for the purpose of creating variability to improve generalization. The main advantage of sample space data augmentation techniques is that they are more intuitive and can be designed to specifically generate desired augmentations for a particular task using simple transformation operations. These transformations can operate on the entire image as a whole, or can be applied in a patch-wise manner, where the input image is first subdivided into smaller regions before applying transformation on designated or random patches. A number of dedicated libraries exist for performing image augmentation operations on input samples. Examples of such libraries include Augmentor [43], Imgaug [44] and Albumination [45]. In addition, many open source (e.g. Octave [46], MIPAV [47]) and propriety software tools such as LabView [48], Matlab [49], Mathematica [50] and ERDAS Imagine [51] have rich sets of in-built functions for image transformation tasks. Common deep learning frameworks such as Caffe [52], Pytorch [53], MXNet [54], TensorFlow [55] and Keras [56] generally include image processing functions to perform image augmentations.

Traditional image and video editing tools also provide utilities for performing various kinds of geometric and photometric transformations. However, performing image augmentation by this process is manual and much more tedious as compared to using deep learning frameworks or specially-designed code libraries such as Augmentor [43], Imgaug [44] and Albumination [45]. Because of this limitation, image and video editing tools are rarely used in practice. Many recent deep learning networks embed transformation operations directly into learning process, modifying the training input data at every epoch. This reduces the amount of storage resources required to implement data augmentation.

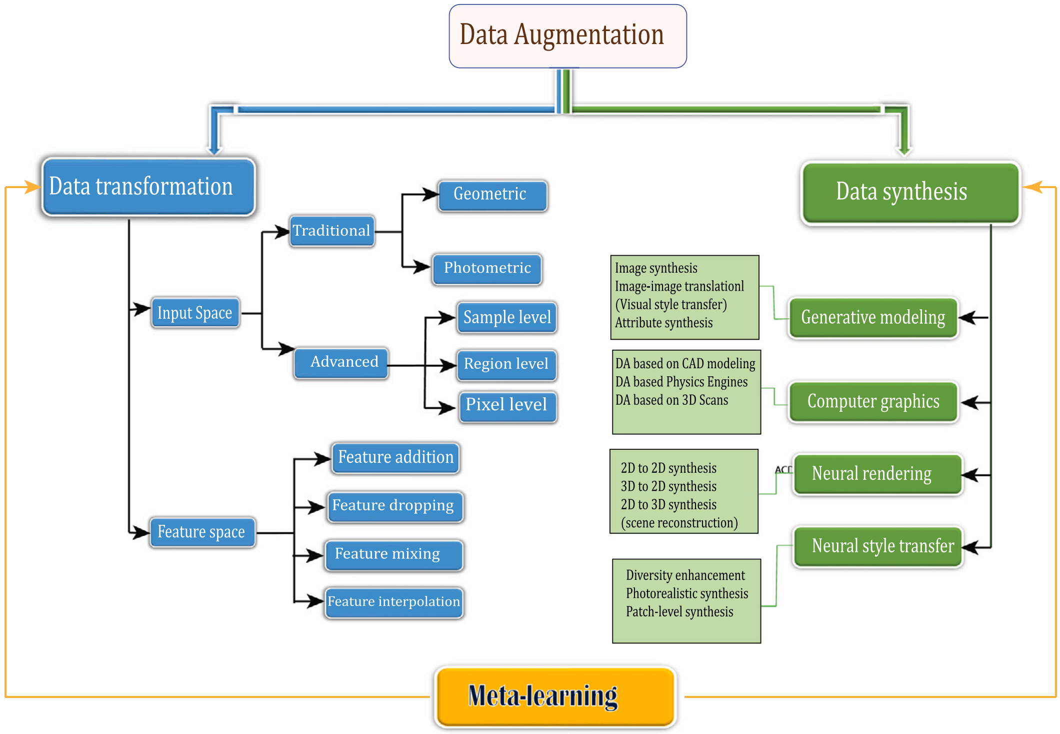

Fig. 2. Taxonomy of data augmentation approaches used in this survey.

2.1. Traditional data augmentation methods

2.1.1. Geometric transformations

Geometric transformations are a class of image data augmentation techniques that alter the geometrical structure of images by shifting image pixels from their original positions to new positions without modifying the pixel values. These transformations are done to modify training data in ways that better describe real-world appearance changes caused by factors such as viewpoint variations, non-rigid deformations, perspective and scale changes. Geometric transformations can be implemented from first principles using polynomial transformations. The reader may refer to [57,58] for implementation details of these transformations. Here we present the basic principles of operation and common applications. Geometric transformations can be further subdivided into affine and non-affine transformations. Affine transformations can be represented by a simple generalized first-order linear equation that requires only few parameter changes to accomplish various specific augmentation operations. Non-affine transformations typically are obtained by using more complex analytical relations involving higher-order polynomial functions.

(a). Affine transformations

The simplest and most commonly used geometric transformation methods are affine transformations — specifically, rotation, shearing, translation, scaling or resizing (without zooming or cropping), mirroring, reflection or flipping. Zooming and cropping are common ways to scale images but they are strictly not affine. Rotations, reflections and translations are a subset of affine transformations known as Euclidean transformations [59]. Despite the simplicity of these techniques, several works [60,61] have shown that they are highly effective in many computer vision tasks. Because of this simplicity and effectiveness, they are often applied as the first course of augmentation before applying more advanced methods [62].

(b). Non-affine geometric transformations

This class of transformations provide a means to simulate more complex geometric artifacts. In many situations, for example, in medical imaging [63] and document analysis [64], complex transformations are required to simulate the intricate geometric changes inherent in the application settings.

Common types of non-affine transformations include projective or perspective transformations (e.g., [65,66]) aimed at introducing new samples with different viewpoints or perspectives of an observer. This typically involves mapping points of the original image to corresponding points in another reference frame according to a pre-defined relation. Data augmentation techniques based on projective transformation are commonly used to extend models trained on canonical image datasets to handle geometric distortions and deformations that occur with wide field of view cameras typically used in overhead imaging applications such as satellite image recognition and UAV (UAV) surveillance, as well as omnidirectional 360° field-of-view (FoV) systems (e.g., in [65]).

Nonlinear deformation [63,64,67] is another important non-affine transformation commonly used to augment data. This transformation can be simulated by introducing more degrees of freedom to allow the basic (e.g., affine) geometric transformation strengths to vary in magnitude across different local image regions. Nonlinear deformation is particularly useful for compensating for appearance variations resulting from natural phenomena (e.g., nonrigid body deformations) and image artifacts caused by various factors (e.g., lens distortion).

2.1.2. Photometric transformations

Real-world computer vision tasks can be adversely impacted by different photometric effects resulting from camera artifacts and shooting conditions (e.g., motion blur, optical noise and distortions), color artifacts and image data corruption. These effects are successfully simulated by photometric transformation methods based on well-known digital image processing techniques, allowing to obtain training images that facilitate accurate and robust visual understanding. Photometric image augmentation methods, in contrast to geometric methods, modify the pixel content of images while preserving their spatial structure. Many of these techniques typically change the composition of RGB channels according to specified functions. Practically, they are implemented using various well-established classical image processing

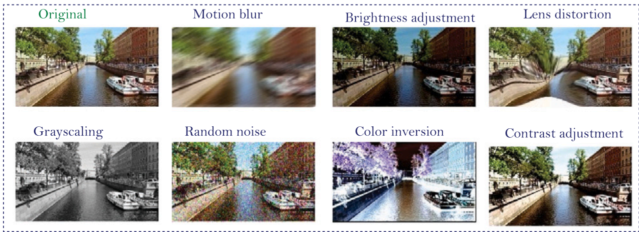

Fig. 3. Examples of the effects of common photometric data augmentation methods.

functions. At the intuitive level, photometric transformation is performed by changing the visual properties of the image, such as its color, brightness, sharpness, contrast and saturation level. Photometric data augmentation can also be carried out in order to enhance the robustness of deep learning models to account for image variations that arise as a result of the influence of environmental conditions such as variations in illumination, weather, times of the day, as well as artifacts introduced by imaging devices, noise and camera settings. The most commonly used photometric data augmentation approaches include color jittering, color space conversion and image enhancement and distortion techniques. The visual effects of different photometric transformations are shown in Fig. 3.

2.2. Advanced input space data augmentation techniques

In this sub-section, we present a detailed discussion of more modern, less conventional input space data augmentation methods. The first class of methods involves learning augmentation operations directly from data. With the second class of methods, transformation operations are applied on training samples in non-traditional ways (typically transforming only sections of input images or mixing image pixels) to generate new samples.

2.2.1. Deeply learned transformation operations for sample-space data augmentation

Owing to the enormous success of end-to-end deep neural networks in solving complex machine vision problems, deeply learned data augmentation methods have recently been given significant attention from the perspective of architectural and design innovations. Many works [68–70] have proposed to incorporate specialized functional sub-models within conventional feed-forward deep learning networks to perform various transformations on input data. The basic idea is to learn effective image transformations to generate additional data in intermediate CNN layers in a way that reflects natural variations of data in the target domain. The additional data generated are seamlessly processed by the deeper layers, leading to an increase in diversity of learned representations. The spatial transformer network (STN) [68], proposed by Jagerberg et al. is a particularly popular approach for learning data augmentations. There are several works that use this method to augment data in many different computer vision tasks (see, for example, [71–73]). In these works, transformation parameters learned from input data are used to warped image samples based on pre-defined affine transformation operations. Mounsaveng et al. [71] used an STN sub-model to learn useful transformed images in an AutoML-based data augmentation pipeline, where deeply learned transformations are used to enrich manually-composed augmentations. Vu and Huang in [73] propose to deeply learn effective transformation operations to warp training samples in an intermediate pipeline of an object detection model.

Jena et al. [74] incorporated an STN sub-model in a data augmentation framework, and used a GAN to learn the distribution of transformation parameters instead of learning probability distribution over input transformations (Fig. 5(b)). They showed that the approach

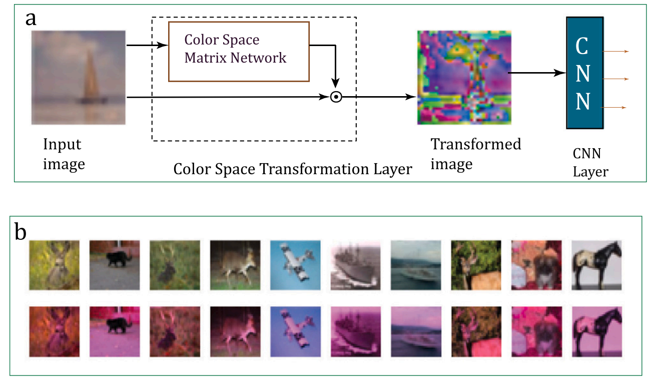

Fig. 4. a. Color Space Transformation Network proposed in [69]. b. Examples of color transformation on sample CIFAR-10 images by the network (also provided in [69]).

is more computationally efficient than learning distributions over input space.

In [69], Karargyris proposes Color Space Transformation Network (CSTN), a simple convolutional network architecture that incorporates a so-called Color Space Matrix Network sub-model to learn useful color space parameters from data and then apply these parameters on the input samples during the training process. The basic structure of the network is shown in Fig. 4. The proposed model increases classification accuracy from 50% to 68% when trained on CIFAR-10 dataset.

Another approach is to use multi-branch networks, where each branch performs a specific, pre-defined image transformation operation. Multi-branch networks have already proven effective in synthesizing more robust representations in computer vision tasks [20]. Consequently, a number of works employ multiple deep neural network branches within an end-to-end deep learning framework for image transformation in the context of data augmentation [74–76]. Using the approach, complex deep learning pipelines incorporating specialized sub-models and additional data paths have been developed to provide more effective data augmentation. The main advantage of the approach is its ability to optimize for specific image transformation operations in each branch separately and then aggregate all the resulting augmented samples as input data in subsequent layers. Zhang et al. [75], for example, proposed a data augmentation scheme that employs three different network models in separate deep neural network branches, where each branch applies a specific image transformation to the input data and the results from the different branches are aggregated and used in addition to the original training data to improve generalization performance (Fig. 6). To perform augmentation, each of the input images is first resized differently by a distributor model and given to different branches consisting of Spatial Transformer Network (STN) [68] sub-models which then apply appropriate transformations.

2.2.2. Region-level data augmentation

Region-level data augmentation strategies typically apply image transformation techniques on image patches instead of uniformly transforming the entire input image. Several studies have shown that these approaches can be more effective in domains such as object detection in crowded settings or visual tracking, where occlusions are usually common. The aim of these strategies is to prevent overfitting of deep learning models to specific features in some localized sub-regions of the training samples by changing their locations or content in different ways. The different region-based augmentation strategies are discussed in the following sub-sections.

(a). Region Deletion

Region deletion or elimination is a type of information dropping technique similar to dropout regularization. Unlike dropout, where the elimination is performed at the feature level, in region elimination, the

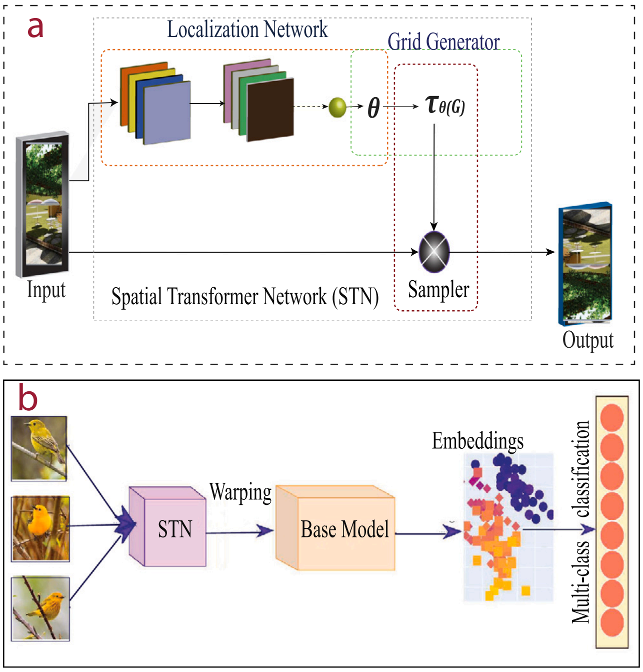

Fig. 5. Basic structure of Spatial Transformation Network (STN) [68]. Implementation of STN-based learned augmentation proposed in [74].

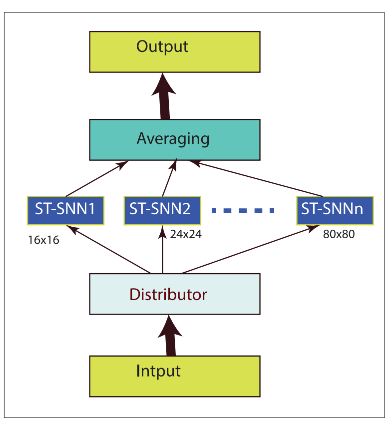

Fig. 6. Multi-column spatial transformer convolution neural network FMC-STCNN) proposed in [75] for performing augmentations.

deletion is applied at the input level. region elimination approaches such as [77–81] delete some random regions from the input samples in the process of training. The idea is to introduce diversity by varying the visual features present in different training samples. Region elimination can also serve as an effective way to simulate occlusion conditions in computer vision tasks [77,81]. This is particularly useful in visual tracking applications [82], where tracked objects are often subject to occlusion as a result of their interaction with other objects or the dynamic environment. Information dropping may also be aimed at eliminating irrelevant (in terms of the particular context) features while preserving more useful visual features. The technique may also be beneficial in situations where appearance information is less important for the recognition task. For instance, Huang et al. in [79] propose a region deletion method for human pose estimation tasks, called Augmentation by Information Dropping (AID), to drop appearance information in images to enable the underlying model to focus more on constraint cues. The authors demonstrated that in crowded scenes where appearance features are usually inaccessible, de-emphasizing appearance cues and learning constraint cues can significantly improve performance.

Cutout [77] deletes random patches from each set of training sample, replacing pixels in the deleted regions with pixel values of zero. This is accomplished by using a square matrix of constant weights to mask out different locations of input samples. The authors showed that the approach is also effective when used in conjunction with other regularization methods such as dropout. Random Erasing [81] proposes an approach that applies randomly-sized masks on random locations and fills into the deleted regions random pixel values between 0 and 255. Instead of applying this transformation on all input samples, in [81] the augmentation are conditionally applied with a specified probability. Additionally, the ratio of the masked area to the total image area, as well the ratio of their aspect ratios are also specified as model hyperparameters. One of the demonstrated strengths of the approach is its ability to model partial occlusion conditions. Yang et al. [78] propose Region-aware Random Erasing to extend the concept of Random Erasing to object detection tasks. The authors developed a so-called Range-aware Random Erasing technique that prevents the square mask of [81] from masking out useful information such parts of the bounding boxes.

Unlike CutOut [77] and Random Erasing [81] that delete a single contiguous region from training images, some new approaches such as [80,83–85] delete multiple, smaller square regions. For instance, Hide-and-Seek (HaS) [80] partitions image samples into smaller grids and then randomly deletes the pixel content of selected grids, leaving those regions with pixel values of 0.

A common limitation of region deletion approaches is the random nature of the deletion process, which does not allow for selecting more appropriate regions for deletion. This can sometimes result in the most useful features being deleted, and thereby negatively affecting performance. To overcome this limitation, Gong et al. [86] proposed KeepAugment to prevent useful information from being dropped in the process of augmentation. During training, the approach first identifies more informative image regions by means of saliency map-based importance ranking of various image parts (rectangular regions). Subsequent processing (i.e., application of augmentation operations) preserves these useful areas while augmentating less useful regions. The approach can be used in conjunction with other region deletion methods like CutOut [77] and Random Erasing [81] to ensure that useful parts of images are preserved in the process of augmentation. Following a similar concept as KeepAugment [86], Chen et al. in [83] proposed a so-called GridMask method. The algorithm first identifies uniformly distributed image regions and then composes these regions into a grid. Thus, the pixel distribution in the image is used as a basis for information dropping. The authors demonstrated, through experimental comparisons, that this approach can outperform more advanced augmentation methods such as AutoAugment [87]. Feng et al. in [84] introduced GridCut which uses a similar region deletion concept as GridMask [83] but also includes a mechanism to adaptively modify each of the grid masks independently based on the specific characteristics of the training images. This produces more diverse, non-uniform masks which in turn provide more nuanced representation ability. In contrast to the above approaches that eliminate whole image regions using rectangular masks, FenceMask [88] generates more sparse, finer grids in the form of wire fence. It reduces information loss and provides better opportunities for generalization. This is particularly useful for fine-grained recognition tasks. Examples of the visual effects of common region deletion methods are shown in Fig. 7.

(b). Region replacement and swapping

Some region level augmentation approaches (e.g., PA-AUG [89], InPS [90] CutBlur [91], CutMix [92], PatchShuffle [93] and Cut-Paste [94]), instead of entirely deleting image regions, crop out patches from training samples, optionally transform them in various ways, and paste them randomly onto different positions of the original image or into other image samples. A major difference between this

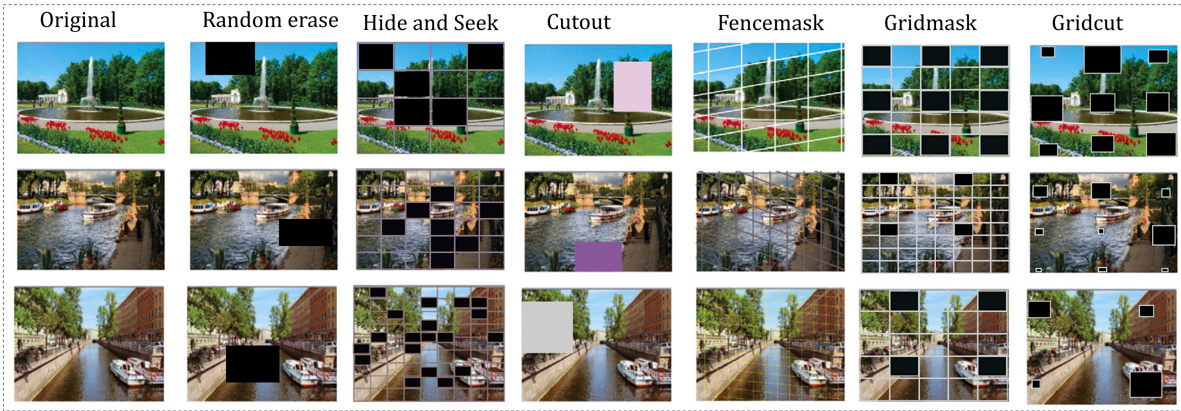

Fig. 7. Visual effects of various region deletion methods.

class of approaches and region deletion methods such as Cutout [77], GridMask [83] and Random Erasing [81] is that the former family of approaches superimpose patches containing the content of other training images while the latter class of approaches create patches with random, zero-out or uniformly filled pixel values. The motivation for replacement methods is to mitigate information loss problems caused by deleting informative content from image samples. In [92], the authors introduced a technique that replaces selected image regions of a given image with a small patch sampled from another image with a corresponding mixing of the labels of the parent image pairs of the given patches. Cut-Thumbnail [95] replaces a random region of an image with a resized (i.e., a thumbnail) version of the same or a different image. Specifically, in the work, the authors propose three different replacement methods: Self Thumbnail (ST), where a resized version of the original image replaces a random region of the same image; Mixed Single Thumbnail (MST), where the thumbnail of another image replaces a region of the image; and Mixed Multiple Thumbnails (MMT), which involves pasting multiple thumbnails onto a single image. Experimental results show that the MST method provides the best performance boost across different datasets. It outperforms state-of-the-art augmentation methods such as CutMix [92], CutOut [77], MixUp [96] and AutoAugment [87]. Resizemix [97] also explores a similar concept of resizing and mixing images. Patch Gaussian [98] replaces a random region with a patch that is derived by injecting Gaussian noise to deleted random image region. The method aims to simultaneously improve both the robustness and generalization accuracy. Yoo et al. in [91] propose to swap high-resolution image patches with their low-resolution versions so as to encourage the resulting model to learn to handle input images of different resolutions. Common examples of region replacement and swapping are shown in Fig. 8. Even though region replacement approaches such as CutPaste [94] and Cut-Thumbnail [95] largely alleviate the information loss problem of previous methods as described earlier, significant information loss may still occur when an uninformative random region (e.g., background) is cut and pasted on a region with more useful semantic content.

Instead of extracting image patches from random local regions and pasting on some other random locations, a number of new works (e.g., AttributeMix [99], FocusMix [100] and Attentive CutMix [101]) have introduced techniques to extract and paste patches based on their semantic content. FocusMix [100] is based on the assumption that the central parts of an image usually contain more relevant information than the margins. The approach extracts patches from central areas and paste on areas around the margins. The authors showed that this outperforms conventional cut-and-paste methods like Cutmix. Attentive CutMix [101] leverages attention maps from intermediate deep neural network layers to identify the most important image parts. The goal is to ensure that more discriminative regions are not adversely affected by the region replacement process in [92]. InAugment proposed by Arar et al. in [102] aims to exploit the natural phenomenon that repeating patterns of many useful features exist at the image level [103] to artificially increase feature diversity of each image instead of generating many new samples. To achieve this, the authors propose to copy small patches from random image regions, apply different scaling transformations to the copied patches and again paste them on some random regions of the same image. This is to ensure that useful information about the image is captured at different resolutions, thereby promoting invariance with respect to scale transformations.

(c). Region combination

Region combination approaches typically add the contents of patches from the same image or from different samples together. This differs from previous approaches that replace (e.g., [94]) or swap (e.g., [91]) different regions. In this case, the goal is to either combine the content of patches pixel-wise (e.g., [104,105]) or compose new samples by joining together smaller patches from different images (e.g., [106–108]). While MixUp and its derivatives implement mixing on the entire samples space for the selected images, region-based approaches perform image mixing at the patch-level rather than on the entire image samples. GridMix, introduced in [84], utilizes the concept of Cutmix method to interpolate image pixels from different source images. Random Image Cropping and Patching (RICAP) [106] randomly crops and stitches together image patches from different locations of randomly selected images from a training dataset (see Fig. 10). The authors in [106] also propose to mix the class labels of source images. This approach provides the benefit of regularization methods like label smoothing. Hong et al. in [108] solves a class-imbalance problem in object detection applications using patch combination method. They composed new samples of scarce instances by cutting and combining patches from different images containing the relevant objects. The approach has shown promise in rebalancing class-imbalanced datasets.

Recently, more advanced patch-level combination approaches (e.g., SaliencyMix [105] and PuzzleMix [104]) employ strategies that allow to identify and utilize more useful information from the given images. Kim et al. [104] argue that conventional sample space mixing approaches that introduce diversity in a random way without taking into account image structure and statistics may potentially lead to confusing training samples, thereby adversely affecting network performance. Also, because the weighting coefficients for image mixing methods are usually fixed and are applied uniformly over the entire image, less important image regions such as backgrounds may be emphasized while semantically useful regions are suppressed. To overcome these limitations, they proposed a blending technique, called PuzzleMix, that conditions the sample mixing process on region saliency and local statistics of the corresponding image samples. This maintains salient information while also preserving the structure and statistics of training data which leads to increased generalization performance. Uddin et al. [105] proposed a model that first learns saliency maps about

Fig. 8. Region replacement and swapping methods.

Fig. 9. Visual effects of different region combination methods.

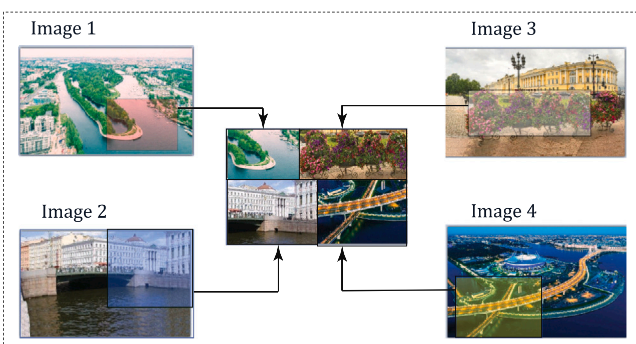

Fig. 10. Basic operation of Random Image Cropping and Patching (RICAP) proposed in [106].

the given images to allow mixing only semantically useful information. Similarly, SuperMix [109] employs a region mixing strategy based on region saliency. Fig. 9 shows the visual effects of popular region combination methods. We summarize the main approaches in Fig. 11.

(d). Pixel-level data augmentation

Pixel-level data augmentation approaches are aimed at generating new samples by performing pixel-wise transformations on the entire image content. These techniques are generally based on the so-called image mixing methods such as MixUp and its derivatives such as [96,110], SamplePairing [111], SmoothMix [112], mWh [113] and AdaMixUp [114]. They commonly blend the contents of two or more image samples together. This is achieved by performing a pixel-wise combination – usually by a form of weighted averaging – of the intensities of the selected images to create artificial ones. The methods can also apply various photometric transformations to the pixel content (e.g., [112]).

Pixel-level data augmentation methods are similar in principle to photometric techniques. However, unlike classical photometric methods, these approaches do not aim to achieve label preservation or provide semantically meaningful representation of images. MixUp [96], a representative work in this area, proposed by Pang et al. applies random weighting coefficients to average pairs of images and their respective labels. Similarly, SamplePairing [111] averages corresponding pixel intensities of a pair of random images selected from the training dataset to generate new training samples. Unlike Mixup [96] that also include label mixing, they retain the label of only one of the image pairs. Lin et al. [115] applied image mixing method known as RoiMix to combine region-of-interest (ROI) proposals from different images for the challenging task of underwater object detection. In [116] Hendrycks et al. propose to first apply various image-level augmentations separately using both geometric and photometric transformations and combining the resulting set of augmented images into a new data sample using element-wise pixel combinations. Image mixing methods such as MixUp [96] that perform pixel-wise combination of patch contents have been shown to be more robust against adversarial attacks [117], while patch approaches that combine individual patches show better generalization performance at the expense of adversarial robustness.

Noisy Mixup and Random Pixels – both augmentation strategies proposed in [118] – are two different pixel-level data augmentation techniques that have also shown promise. Noisy Mixup implements a per-pixel sample combination as in Mixup. Additionally, for each mixing operation, noise is added to randomize the mixing coefficient. Given a set of training images, Random Pixels randomly samples a pixel from one of images for each pixel position.

Lee et al. [112] observed adverse effects of strong edges on model performance in approaches employing image combination techniques, and propose to minimize these effects by using smooth-edge filter masks to reduce the prominence the boundary of images being combined. They suggest an approach in which randomly selected image pairs are first multiplied element-wise by smooth masks before the resulting augmented images are then summed in an element-wise manner. To implement the smoothing effect, conventional 2D square matrices as well as circular masks generated by a Gaussian functions can be used. Pixel-level augmentation strategies are particularly beneficial in countering adversarial attacks. Because of this property, many works (e.g., [98], PuzzleMix [104] and Co-Mixup [119]) have proposed to utilize elements of both pixel- and region-level methods to improve both adversarial robustness and generalization accuracy. Examples of common pixel-level augmentation approaches are shown in Fig. 12.

A major problem with image mixing is the tendency to introduce artificial, out-of-distribution samples to the original dataset. To address this problem, a number of works have proposed to incorporate label-preserving constrains to ensure that the generated samples do not deviate markedly from the distribution of the original training samples. The authors in [96], for example, introduce convex linear interpolation

Augmentation	Basic techniques	Advanced extensions
Region deletion	Deletes a continuous region at a random location from an image – e.g., CutOut, Random Erazing (RE)	Provides a means to preserve useful & informative image regions – e.g., HaS, KeepAugment, GridCut, GridMask
	Deletes randomly sized, smaller patches from random locations of designated image samples	
Region replacement & swapping	Replaces or swaps content of random image locations with informative content taken from same or different image – Cut-Thumbnail, Patch Gaussian, CutPaste	Use of additional mechanisms to identify and replace the most appropriate patches – InAugment, FocusMix, AttributeMix
Region combination	Combines contents of designated image patches to create variability. Patches are stuck together or combined pixel-wise – RICAP, GridMix, CutMix	Provides adaptive mixing mechanisms that determine the most useful mixing ratios based on region saliency – e.g., PuzzleMix, SaliencyMix, SuperMix

Fig. 11. A summary of region-level data augmentation techniques. Each of the three categories of approaches includes basic methods as well as more advanced techniques that perform task-specific augmentations, taking into account context sensitive information.

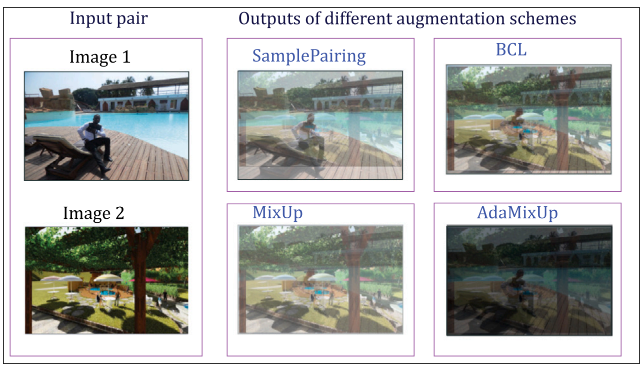

Fig. 12. Examples of pixel level augmentation methods. Note that Mix-up is the same as Between Class Learning (BCL) except that images in BCL are strictly from different classes. Note also that Sample Pairing (SP) and MixUp are same except that SP uses 50%–50% content from each of the constituent image pairs while MixUp can adopt any mixing ratio. Here, we used 35% of Image 1 and 65% of Image 2.

mechanism to provide this constrain. Tokozume et al. [120] employ a so-called Between-Class (BC) learning technique, a method originally devised for the task of combining sound signals [121], to impose constrains on the distribution of the generated samples.

3. Feature space data augmentation strategies

An increasing number of works (e.g., [122–125] are aimed at expanding training data by increasing the diversity of extracted deep feature maps. A common way to obtain diverse feature representation is to directly manipulate feature vectors in intermediate deep neural network layers. These approaches have been inspired by previous research (e.g., [126]) that convincingly showed that lower-level features extracted from deep layers can be used to reconstruct higher-level representations of the input images. In essence, the goal of augmentation strategies following this line of works is to transform data in the feature space in ways that encourage the resulting deep learning model to learn representations that are invariant to image transformations in the input space. Approaches that follow this line of work are particularly attractive because of the fact that manipulating feature vectors or generating new features is much easier and computationally cheaper than generating or manipulating images. Moreover, perturbation at the feature level allows much smoother transformations than dealing with images at the input space. Importantly, feature space approaches can be used to complement traditional data augmentation strategies to improve performance. A major drawback of this approach is that feature variability is produced by perturbations which are not based on domain knowledge and hence, may potentially miss important domain attributes critical to some tasks. Several different techniques can be used to accomplish feature space data augmentation. We discuss the approaches under three broad categories: feature transformation, elimination and combination (mixing).

3.1. Deep feature transformation

An indirect way to obtain diverse features is to manipulate images at the input (i.e., image) level. These manipulations modify specific image attributes that results in corresponding changes in feature representation. Such approaches can generate more useful features to aid performance. In this sub-section, we discuss approaches that directly manipulate existing deep feature vectors [124,125,127,128]. The general idea is to transform extracted feature vectors to enrich representation in feature space while preserving class labels at the sample (i.e., input) level. The objective is to increase diversity of extracted features so as to prevent the underlying model from learning specific configurations of the input data. Shen et al. [127], for example, propose to perform random affine operations – specifically, translation, scaling and rotation – on learned feature maps to enhance their variability. Li and Liu [129] propose to inject adaptive Gaussian noises into some deep CNN layers of deep neural networks to encourage feature diversity and prevent overfitting. Gastaldi in [128] proposes to replace summation operations that are typically used to combine features from multi-branch networks with affine combinations — a technique where stochastic affine transformations are applied in the summation process. The technique was used to improve performance of ResNeXt [130] models on several different computer vision tasks. A more recent technique, Shakedrop [131] extends this concept to several different multi-branch CNN architectures. KRAM, proposed by Jia and Zhang in [125], employ k-nearest neighbor (kNN) technique to exploit the feature relationships among instances in multi-dimensional classification setting, and consequently transform the original feature representations to improve the performance of multi-dimensional classification models. The authors in [132] describe a feature transformation approach known as Linear Delta. It consists in first computing the difference between a pair of feature vectors extracted from two training samples and then transforming a third feature vector by adding this difference to it.

3.2. Feature addition and mixing

Another line of research is aimed at enriching learned features by augmenting them with artificially generated feature vectors (e.g., in [133,134]) or combining features extracted from different input samples (e.g., [99,135]). Smart Augmentation [99], for example, introduces a data augmentation method that involves mixing features of different input samples. In [136], Faramarzi et al. introduced Patchup, a feature mixing method that uses 2D filters to mask out feature blocks in the intermediate layers of a deep neural network. The feature blocks are later mixed or exchanged with corresponding feature vectors from

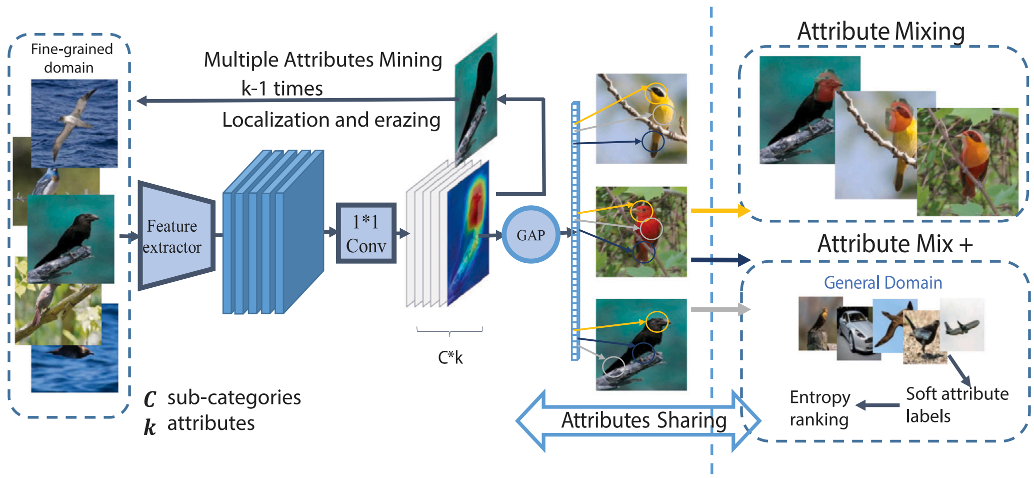

Fig. 13. Architecture of AttributeMix proposed in [99].

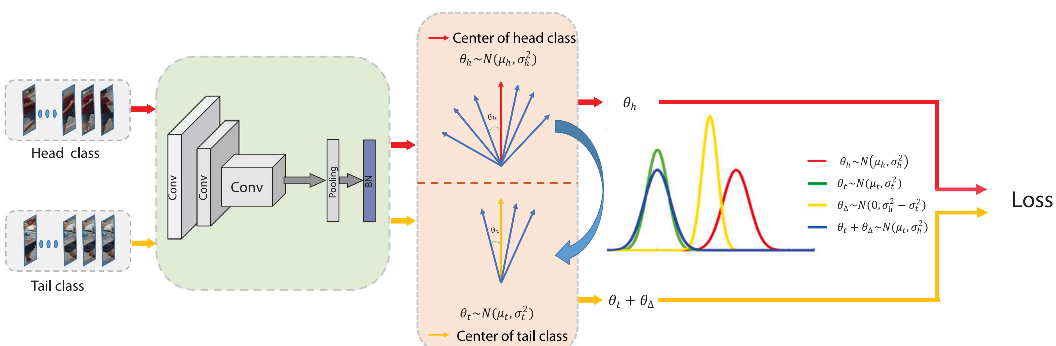

Fig. 14. Deep feature augmentation scheme proposed by Liu et al. in [124] to alleviate data imbalance problem. Data augmentation is accomplished by first calculating the angular variances of head classes and applying this information to perturb tail classes in the feature space, thereby generating rich features for the tail classes.

different training samples. Li et al. [135] propose to learn and mix image attributes at the feature level for fine-grained bird classification (Fig. 13). Their approach, known as Attribute Mix, first learns semantically useful attributes from different sub-categories with image-level annotations using a dedicated attribute learning sub-network. Different image features extracted from same attributes are then mixed by a so-called Attribute Mix technique proposed by the authors to generate richer attribute-level features that are shared among the different images.

Interpolation methods [137–140] have also been widely used to help generate additional features to augment data in feature space. The methods generate additional features based on the relationships between existing neighboring features. This, thus, leads to extra information that fills potential gaps in the representations. For instance, Zhen et al. [140] propose to both expand and contract learned features using interpolation methods. In [137], DeVries and Taylor use both interpolation and extrapolation techniques in addition to noise perturbations to further increase the diversity of extracted features. Li et al. [138] exploit moment information of input image pairs from extracted CNN features and interpolate the corresponding labels in feature space to generate more diverse feature representation that improves robustness in object recognition tasks.

Some feature-level data augmentation techniques (e.g., [124,141, 142]) have been proposed to specifically tackle imbalance data problems by transferring feature statistics from classes with many training examples to those with few examples, with or without additional transformations. They estimate the statistics of samples with large numbers based on the characteristics of their feature distribution in feature space and then transfer this information to similar classes with less examples. These approaches are based on some simplifying assumptions about the distribution of feature statistics such as having a Gaussian distribution and that the mean and variance of features from similar image classes are also similar. For instance, Liu et al. [124] employed this approach to tackle long-tailed distribution problem in object re-identification tasks. To simulate long-tail problem settings, they first converted a large-scale person re-identification dataset – specifically, the DukeMTMC-reID dataset – into a long-tail dataset by reducing the number of samples in some classes. They then computed the angular variances of head classes and applied the information about the angular variances to perform local perturbations of the artificially created tail classes in the feature space, thereby generating rich feature sets for the tail classes. The functional diagram of the approach is shown in Fig. 14. Similarly, Yang et al. [141] applied feature space augmentation technique based on the statistics of deeply learned features to re-balance an imbalance dataset for few-shot image classification tasks. In [142], the authors interpolated between data points of minority classes in order to simulate additional samples for these classes.

Instead of combining deep features from different training samples, some approaches (e.g., [134,143–146]) propose to enrich feature sets in deeper layers by leveraging information (i.e., extracted features) from multiple layers for the target category itself. For instance, Khan and Fraz [143] developed a technique that combines both low- and high-level features extracted from deeper CNN layers to create augmented features for the purpose of data augmentation. Also, Manifold mixup [147] combines features from multiple deep network layers and propagate them further for re-sampling. Similarly, Hsieh et al. [144] exploited deep features at varying semantic levels to enrich feature representation. PointMixup [148] uses interpolation technique to combine different levels of point cloud feature values in intermediate layers. Konno and Iwazume in [149] proposed a more simple approach that involves extracting features from the last but one layer of a deep CNN model which are then used as semantic input to independently train the final layer. They showed that this simple technique can create the appearance of wider feature diversity which can indeed improve generalization in at least some scenarios.

In contrast to approaches that directly utilize deeply learned features extracted from different layers for augmentation [143–146,149], some works [124,133,134] aim to artificially generate additional features and apply them together with learned features for training in subsequent layers. For example, Bello et al. [133] utilized a Transformer-based DCNN model to implement a self-attention sub-network that allows to generate features independent of convolutional operators. They then augmented convolutional features with the self-attention-generated features. They demonstrated the effectiveness of the approach through extensive tests on different deep learning architectures on image classification and object detection tasks. In [134], Wang et al. performed feature level augmentation by randomly sampling feature vectors from a zero-mean normal distribution obtained with the help of pre-computed category-specific covariance matrix of extracted features. The randomly sampled features for each class are then applied to the corresponding category feature maps in deeper layers of the network to enhance performance.

3.3. Feature elimination

Feature elimination is a well-known class of regularization techniques for improving deep learning models that operate by eliminating features in intermediate CNN layers. Among this family of approaches, Dropout and its derivatives such as [11–13,150,151]have been particularly popular and have been used in state-of-the-art CNN backbones models such as VGG [152], Wide ResNet [153], Shake-Shake [128] and PyramidNet [154]. By eliminating neural connections, the approaches introduce sparsity into the network. The literature in this area is quite extensive to be fully covered here. Therefore, we mainly focus on some of the most promising developments in this area. Traditional feature elimination approaches [12,13,150,151] generally involve randomly deleting a subset of features and their associated synaptic connections from intermediate layers. This is commonly achieved by specifying a probability P with which individual perceptrons are dropped out of a base network in the course of each training epoch. This is equivalent to training different variations of the base network. The resultant model, after training, averages the effects of the different model configurations. Consequently, the approach encourages diverse feature encoding and prevents overfitting to training data. This advantage partially stems from the fact that the network learns to avoid co-adaptation [13] by decoupling strict internal structure from semantic content. Apart from improving generalization accuracy, the approaches have been shown to improve robustness against adversarial attacks. They are also are useful for handling challenging visual recognition tasks such as severe occlusions in real-world scenarios [155].

In the context of data augmentation, dropout and related techniques can be viewed as a feature space version of the information dropping (region deletion) methods covered in Section 2.2. They can be combined with other regularization methods to improve the performance of deep learning models. Several studies [7,128,153,156] have, however, reported that some regularization techniques, notably Batch Normalization, adversely affects the performance of Dropout techniques in deep neural networks. Newer data augmentation techniques such as [128] have been designed to overcome this limitation and allow dropout and related methods to work in conjunction with other regularization methods. Instead of completely dropping some nodes on each update during training, Shakeout [157] proposes an approach that randomly suppresses some features while enhancing others. Zoneout [158] is similar in principle to Shakeout. Instead of deleting features, it encourages some random units to maintain their original values while the remaining ones are made to update theirs.

Because of the effectiveness of feature space augmentation methods, several researchers [148,159,160] have explored similar strategies for non-standard image modalities for machine perception. For instance, Zang et al. in [159] proposed an approach to drop information from point cloud-based feature maps of deep neural networks. Specifically, the approach employs three sub-models to drop information at various levels in a point cloud network – namely: DropPoint, DropCluster and DropFeat – which, respectively, drop information at the point, cluster and feature levels.

While earlier approaches drop features randomly, many recent works [161,162] are aimed at identifying and dropping features that are considered to be less important for the given computer vision task. The goal of these information dropping approach is to eliminate irrelevant (in terms of the particular context) features while at the same time retaining useful features. Choe and Shim [162] used attention mechanism to identify such perceptually-irrelevant features for dropping. Some new works (e.g., DropBlock [163], Batch DropBlock [161] and SD-Unet [164]) have proposed to rather drop a whole group of units forming a contiguous region instead of individual elements in as in approaches like Dropout [12]. DropBlock [163] is one of the earliest works to introduce a well-structured dropout scheme that is spatially compatible with CNN architecture. The approach can be considered as an implementation of Cutout [77] at the feature level. Like cutout, it applies zeroed-out elements to randomly selected regions of feature maps. The region dropout scheme in SD-Unet [164] has been designed specifically for U-Net architectures [67] to overcome overfitting in medical image segmentation tasks. Dai et al. [161] propose a person Re-identification (ReID) framework consisting of two network branches. One branch is an ordinary CNN that learns global image features while the other branch implements attention mechanism using a Batch DropBlock (BDB) sub-model to perform region-wise feature dropping. The idea is to repeatedly drop units from the same region for each batch of images corresponding to the same human body parts in the input space. This helps to focus attention on the available body parts and thereby encouraging the ReID model to learn these parts.

4. Image synthesis approaches

Synthesizing data from "scratch" is an alternative means to address data scarcity problems in computer vision. The use of artificially-generated data for training machine learning models is useful when data for a particular task cannot be obtained, or is of very poor quality to meet the requirement of the machine learning task. Artificial data can also be deliberately constructed to provide richer representations and better annotation schemes suitable for a particular task. We discuss the most important data synthesis approaches and their specific characteristics and use-cases in this section.

4.1. Computer graphics modeling

Computer graphics modeling represents an important class of data synthesis methods that can be used for training computer vision models. Computer aided design (CAD) tools are able to create complex 2D and 3D models of objects which can be used as training data. Modeling approaches based on modern 3D game engines can synthesize very large dynamic scenes.

4.1.1. Augmentation with synthetic 2D images and static 3D scenes

The basic principle of computer graphics modeling consists in representing scene information using primitive geometry elements such as vertices and corresponding sets of edges that connect them together. Besides the basic geometric entities, functional descriptions of scene parameters such as materials, lighting, textures and camera position are also specified. These elementary components are then composed into realistic, high-level 3D objects. The generated 3D data can directly be used for training models on 3D visual recognition tasks.

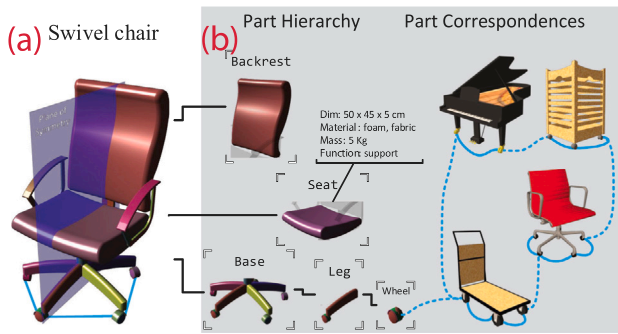

Fig. 15. ShapeNet [165] contains annotations for individual objects constructed as 3D CAD models (a). These objects are built from compositional parts. In addition to providing WordNet definitions for objects and parts, part hierarchy, as well as part-to-part and point-to-point correspondences are specified (b).

Fig. 16. Examples of synthetic scenes from the Virtual Kitti dataset [169]. Image is from the original work that introduced the dataset.

Alternatively, standard 2D pixel images can be generated from the intermediate 3D data through rendering and rasterization. In addition to generating realistic 2D and 3D data, CAD models can also produce 2.5D (RGB-D) images – for example, in [166] – through rendering. Because of their inherent 3D grounding, rendered 2D images generated by computer graphics tools provide better opportunities for applying physically-plausible augmentations. Datasets generated from 3D computer graphics tools provide rich annotation schemes for complex tasks requiring reasoning about objects in terms of their compositional parts (see Fig. 15, for example).

Several large-scale datasets [165,165–167] generated by CAD modeling methods exist for training deep learning models for a wide range of application domains like indoor scene understanding [166,168], person re-identification [167] and 3D object recognition [165]. These datasets support common computer vision tasks such as scene depth estimation (e.g., in [166]), object detection (e.g., [167,167]), semantic and instance segmentation (e.g., [165,166]) and pose estimation (e.g., [166]). Typically, data augmentation approaches based on traditional CAD modeling tools address smaller and more restricted virtual scenes.

4.1.2. Data augmentation based on dynamic 3D scene modeling

Recently, computer graphics modeling techniques based on 3D game engines have expanded the scope of 3D computer graphics tools with regard to data synthesis for training machine learning models. These tools allow larger and more complex and dynamic

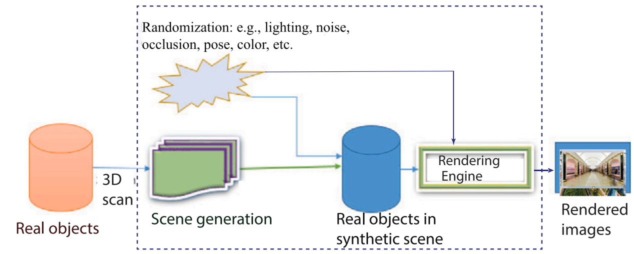

Fig. 17. Simplified pipeline of image synthesis approaches based on 3D-scanning of real objects. 3D scanners are first used to obtain the basic geometrical structure of real objects. Learned or analytically defined scene variables are then manipulated to provide more diverse representations of the object before rendering into 2D images.

scenes with interactive objects to be generated. Datasets produced by such techniques tackle more complex tasks such as augmented and virtual reality [170], autonomous driving [171], and semantic scene understanding in complex environments [169]. This class of computer-generated 3D datasets can incorporate additional high-level information about the underlying objects represented, and allow more complex problems to be solved. For example, computer graphics-based 3D scenes can explicitly specify the coordinates of different objects in a scene. Moreover, in addition to the visual characteristics, they can incorporate real physical properties such as density, friction and mass, and thus enable real-world behavior of objects to be modeled. Popular datasets generated using 3D game engines include Virtual Kitti [169] and Synthia [172] for urban scene understanding and autonomous driving, LCrowd [173] for crowd analysis. Fig. 16 shows samples of realistic scenes from the Virtual Kitti [169] dataset.

4.2. Neural rendering

Neural rendering [174] aims to accomplish the rendering process by using deep neural networks to learn parameters that describe the underlying 3D scene. The process, in essence, transforms image pixels (encoded in deep neural networks as convolutional features) into pseudo-continuous 3D geometry characterized by high-level visual information – including lighting, color, textures, materials, and motion – as observed by the camera. The knowledge about image generation is then exploited to create realistic images by applying physically-grounded manipulations on the constituent parameters of the intermediate 3D scene. The process of obtaining scene information from image pixels, often referred to as inverse graphics, has a long history in computer vision [175]. However, for a long time, owing to computational resource constraints, it was extremely challenging to implement this method using neural networks. Recent advances in computer hardware and machine learning algorithms have now enabled practical realization of the process using only neural networks. State-of-the-art differential rendering techniques allow realistic 3D scenes to be generated from 2D pixel images. Differential neural rendering implements the entire scene generation pipeline in an end-to-end manner. With this approach, a deep learning model is trained to generate synthetic pixel images that are consistent with real-world images by optimizing different parameters related to the constituent objects' shapes, textures, poses, and scene illumination conditions. Large-scale datasets generated using neural rendering methods (e.g., RTMV [176] and Synscapes [177]) are already available for training machine learning models. These datasets generally target complex applications such as outdoor scene understanding and autonomous driving tasks. In some cases (e.g.,[178,179]), real objects are scanned into virtual modeling environments, where their basic 3D representations are further enhanced before rendering. A simplified pipeline of this process is shown in Fig. 17.

Earlier neural rendering methods such as [180,181] utilize explicitly constructed geometric primitives and basic scene parameter within deep learning pipelines to describe 3D scene. This method has the disadvantage of high memory requirements, thereby limiting the resolution of voxels and the ability to handle large scenes [182]. Many modern neural rendering techniques increasingly rely on implicit or learned representations that employ continuous functions to describe the underlying scene elements. Current state-of-the-art involves using so-called Neural Radiance Fields (NeRFs) [183]. The basic principle of NeRFs is based on the idea of light transportation through different points in a 3D space. The representation approach, thus, to some extent, is physically grounded.

State-of-the-art NeRF generators such as GRAM [184] are capable of synthesizing highly-detailed, 3D-plausible and view-controllable data with just a few 2D training examples containing a small subset of views. These models can seamlessly generate new images with the desired appearance properties depicting real-world phenomena like different illumination conditions, poses, colors and textures when provided with a single image. They typically provide explicit mechanisms to adjust scene settings. For instance, the approach in [185] provides so-called appearance codes that can be manipulated to dynamically introduce appearance variations corresponding to different real-world phenomena such as the camera settings, time of day, as well as seasons and weather conditions. Many of the more recent works such as Block-NeRF [185] and Mega-NeRF [186] are designed to generate very large dynamic scenes of outdoor environment in the form of videos from a collection of sparse images. In [185], the authors used Block-NeRF to generate a large-scale virtual environment targeting autonomous driving applications. They divide large scenes into smaller units and then employ multiple networks to process different scene parts. With their approach, only required parts of the scene representing the current "visual range" are loaded as one traverses the scene.

4.3. Generative modeling

In recent years, the generative adversarial network (GAN) [187] has emerged as the most popular class of modern deep learning models for synthetically generating image data. The GAN implements a form of minimax game between two sub-models — a generator which generates data close enough to real data, and a discriminator whose role is to distinguish between real data and the artificially generated. The competition between these two sub-models ultimately leads to the network generating high quality data. The conditional GAN (CGAN) [188] utilizes reference images as prior information to encourage the model to learn specific visual features. Approaches based on the conditional GAN, thus, provide a form of control over the visual properties of the synthesized data. This additional control can be leveraged to specifically transfer the desired visual appearance of one source image to a given target image.

4.3.1. Two-dimensional image synthesis and manipulation

The possibility of using GANs to synthesize new realistic image samples was first proposed by Godfellow et al. [189]. A major limitation of this concept is the difficulty of generating sufficiently large image data suitable for training computer vision models. Radford et al. [190] proposed a better method for of image synthesis using a fully convolutional architecture instead of the fully-connected layers proposed in [189]. Through further refinements (e.g., [191–194]), today, GANs have several exciting uses related to data synthesis and augmentation. In addition to synthesizing image data [194–196], generative modeling methods are also widely used to perform domain adaptation tasks. In this application scenario, the generative model is typically used to transfer the visual appearance of images taken in one domain (or under a set of conditions — e.g., clear weather) to a new domain (or under different set of conditions — e.g., foggy weather [197]). The difficulty of collecting image data in adverse conditions such as snowy, rainy or hazy weather makes it vital to be able to transfer the visual appearance of images taken in normal conditions to such adverse weather conditions. Generative modeling techniques have also proven effective in solving specific image processing and transformation problems. Notably, they can be used to generate clean labeled images from corrupted datasets [198]. GANs have been used to generate multiple stereoscopic views from a single view [199] or different poses from a single pose [200]. The use of generative models for performing specific photometric image operations such as conversion from gray-scale to colored images [201], and geometric transforms [202] is also common. Variational Autoencoders (VAEs) [203], to a less extent, have also been employed to synthesize data for the purpose of extending training data (e.g., in [204,205]). VAEs are based on a combination of Bayesian statistics and deep learning methods. The VAE learns the probability distribution of training data. When trained on image data, for example, knowledge about this probability distribution is used to generate new samples similar to the input data. The conditional variational autoencoders (CVAE) [206], much like the conditional GAN (cGAN)) [188], enables desired output properties to be simulated based on auxiliary input data. Because of the difficulty of achieving high degree of photorealism with VAE models, many recent data synthesis methods tend to favor GAN models. The main strengths of the VAE are that it is easier to train and does not suffer from the problem of mode collapse associated with GAN models. VAEs also provide additional means to flexibly create variability by manipulating variables in the latent space. While GANs generate better quality images, they are generally limited by the level of control that can be exerted on the generation process. Because of this, some state-of-the-art generative modeling techniques (e.g., [207–209] utilize both GANs and VAEs so as to leverage their complementary benefits. Some recent works have proposed to extend VAEs to 3D domains [210,211]. These approaches typically utilize implicit volumetric representations and then employ VAE models to perform various transformations and interpolations in the latent space before re-rendering the final output.

Generative modeling methods can be used independently as an alternative data synthesis method [212]. They can also be used as a refinement step to introduce better perceptual properties to data generated by other synthesis techniques. For example, generative modeling techniques have been used to refine synthetic images generated by computer graphics modeling tools [213,214] or from neural rendering pipelines [182]. This is an important task since it is challenging and time-consuming to create highly detailed and photorealistic images using these methods.

4.3.2. Large-scale, realistic 3D scene synthesis by generative modeling

While generative modeling methods based on GANs and VAEs have been used to successfully transform synthetic 2D image data into realistic-looking data [187,215], conventional adaptation strategies usually apply a single or multiple styles uniformly to the entire scene at a global level. The main limitation of the approach is that the transformations generally fail to provide context-relevant transformations that correctly deals with the specific objects in the scene. Therefore, it has been proposed to rather adapt the appearance and structure of individual objects separately so as to ensure more nuanced and consistent object-level characteristics (e.g., in [216–219]). While these approaches generally handle appearance properties well and guarantee some level of style consistency, they still lack spatial consistency and the ability to manipulate geometric aspects of the 3D scene. Recently, many 3D GAN models have emerged as a workaround [182,220–222]. These techniques commonly leverage the 3D representation power of neural rendering models and the ability of the generative process to simulate statistical distributions of real-world image features. In particular, the approach enables GAN-based stylization to be applied to 3D scenes while maintaining multi-view and spatial consistency. It also offers better controllability of high-level visual attributes by allowing 3D-aware adjustment of low-level scene parameters. In [221], Niemeyer and Geiger proposed a NeRF-based GAN model that decouples the foreground and background objects, allowing them to be manipulated separately in the course of synthesis. It allows visual attributes such as pose, shape, size, color and texture of individual objects to be adjusted independently without affecting other objects' visual properties.

4.4. Neural style transfer

Neural style transfer (NST) [223] is a technique used to construct novel images based on the combination of hierarchically different layers of deep CNN models. Like GAN-based style transfer, the goal is to synthesize images that have the same appearance features as a given reference image but maintains the original image structure.

To enable this approach to be used for solving data augmentation problems, the basic scheme proposed by Gatys et al. in [223] has been greatly extended. These extensions are aimed at addressing various limitations of the original model and provide new capabilities and practical uses. The key aspects of these extensions include: improving the perceptual quality of the generated samples [224], increasing the training speed of the NST process [225,226], providing more control over the generated output data [227], enabling the synthesis of photo-realistic and semantically meaningful images [228,229], and increasing the multiplicity of styles that can be applied simultaneously on a given content image [230].

To improve the perceptual quality of output images, Li et al. [224], for instance, proposed an auxiliary refinement step based on the application of whitening and coloring transform (WCT) to manipulate the extracted content features in convolutional layers according to the statistics of the style features. While the approach in [224] was aimed at eliminating image distortions and enhancing the overall photometric quality of generated data, it – like most previous works – was still based on transferring non-realistic, artistic styles. In [231], a photorealism-preserving stylization method is proposed that relies on a manifold ranking process as a smoothing function to eliminate noise that often accompanies previous stylization procedures. Luan et al. [228] accomplish this by introducing a training loss term that optimizes the generated output on photorealism. In [229], a patch-based style transfer method is used to improve the photorealism of generated image data, while at the same time, increasing memory efficiency in the stylization of high resolution images.

4.4.1. NST as a domain adaptation technique

Photorealistic stylization approaches seek to manipulate style and content representations encoded by different convolutional layers in order to generate perceptually meaningful data for training deep learning models. In particular, photorealistic NST stylization allows a specified content image or a set of content images to be styled based on a given high-level semantic visual context. These approaches have been used to address domain adaptation problems, where the NST network is used to extract useful visual features from images with the target appearance and apply them on target images which lack this appearance. For example, the authors in [232] used an NST model to alter images taken in the daytime to the appearance of nighttime images, where labeled data is generally inaccessible. Similarly, Li et al. [231] transferred images taken in clear weather to snowy conditions. More advanced neural stylization approaches [233,234] have been designed to learn the semantic appearance as well as physically-plausible behavior of objects of interest.

4.4.2. NST as a means to prevent bias towards low-level visual features

Since computer vision models trained on large-scale image datasets are often biased toward local object features such as color and texture [235], many style transfer approaches such as StyleAugment [236] and StyleMix [237] are primarily aimed at mitigating this shortcoming by encouraging models to focus more on shape information. StyleMix [237] is a modification of sample-level image mixing data augmentation strategies such as MixUp [96] using the principles of neural style transfer. The authors propose extracting content and style information from input image pairs and then independently manipulating them before mixing. StyleCutMix, also proposed in the same work [237], is a neural style transfer extension of the classical region-level augmentation methods based on cut-and-paste [94]. The authors in [236,237] demonstrate that manipulating the style and content information separately significantly improves generalization performance. The impressive generalization performance is based on the fact that this type of augmentation generates new images with varied distribution of low-level visual features while maintaining the high-level structure and semantic content.

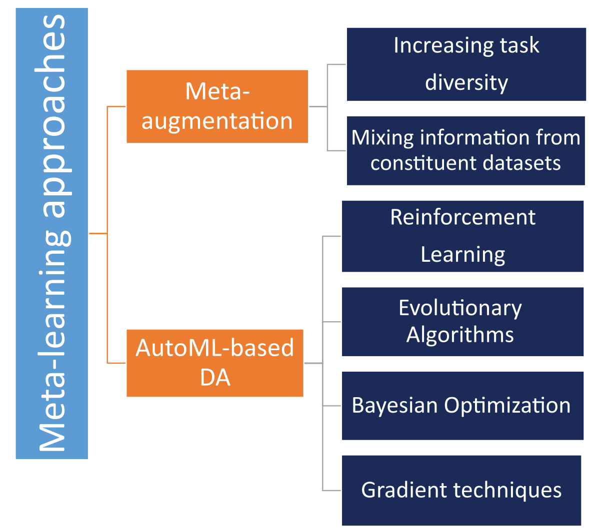

Fig. 18. In the literature, two main conceptions of meta-learning are distinguished: (1) Meta-learning approaches that leverage auxiliary datasets to enable machine learning models to encode general knowledge and (2) approaches that automate the process of data augmentation. Data augmentation in the first case involves mechanisms to increase the diversity of data. In the second case, augmentation is achieved by utilizing optimization techniques to find the best transformation operations and their optimum parameters.

5. Meta-learning approaches

Recently, meta-learning has become a hot topic in machine learning and its application in data augmentation is now widespread, In published literature, two broad conceptions of meta-learning have been advanced:

- Meta-learning approaches in which the underlying deep learning model leverages meta-data to encode generalized representation that can perform well across different tasks. These techniques are also aimed at endowing models with the ability to adapt to new tasks effectively. The goal of data augmentation in this class of works is to employ few-short learning methods to drastically increase sample efficiency of machine learning models.
- Optimization-based methods that focus on automatically learning the best model hyperparameters for generalization. This type of meta-learning typically utilize automated machine learning (AutoML) techniques to find the most effective augmentation operations for a given task. The main goal of autoML-based data augmentation is to automatically generate augmentations and train a deep neural network on the augmented data to obtain the best possible performance for the given task.

In this section, we explore data augmentation in both types of meta-learning settings. The simplified taxonomy of the approaches is presented in Fig. 18.

5.1. Data augmentation in meta-learning settings

Currently, deep neural networks typically utilize large quantities of labeled data to train for specific tasks. It is often expensive and, in some cases, impractical to obtain sufficient data for a target task. Moreover, at inference time, a machine learning model may need to solve many different but related tasks at the same time. Meta-learning, in this context, can be considered as an approach to endow machine learning models with the ability to learn useful concepts that allow generalization of learned knowledge to new tasks. In this subsection, we first look at meta-learning as a data-centric approach to the generality of data representation (meta-data). We then present techniques used to achieve data augmentation in the context of meta-learning.

Leveraging meta-data: In view of the difficulty in obtaining large datasets that are representative of multiple tasks, this concept of meta-learning leverages auxiliary datasets that contain only a few examples per category with prior representation obtained from a large dataset of a base category. Typically, the base model is provided with support data to update the learned representation. Then query input is given which is used to tune the model's generalizing performance on a test set. This procedure is widely used in few-short learning (FSL) tasks [238].

Meta-augmentation: Overfitting in meta-learning can result from training the base model or from the auxiliary tasks [239]. Therefore, a common way of realizing data augmentation, meta-augmentation, is aimed at expanding the support dataset (i.e., the number of samples per class) or the range of tasks (new image classes) or both. Thus, apart from diversifying data for each class, an important objective is to create more diverse tasks for each training sample, thereby increasing model adaptation to new tasks. For example, [240] implements a so-called task-augmentation-by-rotating method that creates additional tasks by applying large rotations to the support dataset and composing them into new classes. Similarly, the approach in [239] proposes to address memorization and learner overfitting by expanding the task distribution in the support set. The authors also suggest that injecting conditional entropy-preserving and conditional entropy-increasing noise could also improve generalization performance.

Another useful way to extend training data is to mine and combine useful information from the base and supplementary datasets to enrich representation. MetaMix, proposed by Yao et al. [241], seeks to average feature vectors from both the query and support sets and use that as additional information to augment the query set in order to prevent the model from memorizing the query data while disregarding the support data. Ni et al. propose to extend the concept of MaxUp augmentation [242] to meta-learning scenarios. Their new approach, Meta-MaxUp [243], iteratively applies random transformations to designated batches of tasks and their corresponding query and support data during each training episode, and then find and minimize training loss for tasks with maximum loss. This is to rectify perturbations that might produce aggressive augmentations which could potentially harm performance.

5.2. Meta-learning approaches based on AutoML

Meta-learning, in the context of automated machine learning, can be applied to both data transformation and synthesis-based augmentation schemes. However, it is most often used to find optimum augmentations in data transformation-based augmentation strategies. Only a few works (e.g., [244–247]) have employed meta-learning on image data synthesis tasks. In the context of data augmentation, this involves constructing basic functions that can be applied to synthesize new data. The goal of meta-learning in this case is to automate the process of applying these basic data synthesis operations to produce the best data for augmentation. This approach to meta-learning relies on automated machine learning (AutoML) techniques. For a given task and dataset, AutoML methods seek to automate the process of selecting the most effective augmentation strategies that provides the best performance. To achieve this, typically, a large set of basic operations with varying magnitudes are defined. Heuristic search algorithms are then employed to select the most effective transformations that produce data augmentations. In the process, different possible combination of the augmentation operations are tested.

While meta-learning-based data augmentation has mostly been utilized with data transformation-based augmentation techniques, it can potentially be useful for data synthesis approaches. Indeed, Sun et al. [248] and Mishra et al. [249] have demonstrated the efficacy of the approach in data synthesis tasks. In this case the meta-learning problem typically consists in automatically optimizing basic operations that generate scene geometry primitives and visual properties (e.g., color, pose, textures and lighting) for the given task. The selection of "best" augmentation strategy is based on validation results on test data that evaluates all candidate augmentations according a pre-defined performance criterion. Thus, the process of searching for effective augmentations is carried out automatically without recourse to prior information about the effectiveness of the augmentation operations. The main tasks in the autoaugmentation pipeline are (1) formulation of primitive augmentation operations to be used for data transformation or synthesis and (2) optimization process to select best performing augmentations for the target task. We review current approaches to accomplishing each of these tasks.

5.2.1. Composition of augmentations for AutoML

The first and most fundamental question in the design of data augmentation is the choice of the types and nature of augmentations to incorporate. Currently, most AutoML implementations tackle this sub-task manually. This problem is solved by considering the specific requirements of the computer vision task and constructing a complete set of transformation operations necessary to carry out data augmentation. Besides the construction of basic image transformation functions to be used for augmentation, the search space design also includes the specification of augmentation policies. Augmentation policy in the context of automated data augmentation is a specification of the sequence of primitive augmentation operations (i.e., image transformation functions), their intensity values and selection probabilities. For Autoaugment [250], the authors defined several different types of primitive image processing operations to transform images in specific ways (e.g., rotations, scaling, blurring, contrast adjustment, etc.). In addition to the specific transformations, the appropriate magnitudes (i.e., the range of scale factors and rotation angles) were also specified. A large number of works on automated data augmentation adopt this process. These include Random Autoaugment (RA) [251], Fast Autoaugment [252] Faster Autoaugment [253], Direct Differentiable Augmentation Search (DDAS) [254], and Patch AutoAugment (PPA) [85]. In these works, fixed transformation operations are constructed with respect to various types of image augmentations.

Recently, a few works [71,255–257] have proposed to compose basic augmentations using learned image transformations. This approach is a transition from using explicit analytical functions that define image transformations to learning more useful, task-dependent transformations directly from training data. In practice, such automatically learned transformations can be obtained using specialized CNN models such as Spatial Transformation Networks (STN) [68] or Deformable Convolutional networks (DCN) [258]. For example, Mounsaveng et al. [257] employ an STN sub-model to generate transformations to be applied in the augmentation process. In order to generate more diverse augmentations, Miao and Rahman [257] combined transformations learned by an STN sub-network with those constructed manually according to the specification in [250]. Smart Augment [135] also learns to compose useful augmentations in the process of training. The approach uses an auxiliary network dedicated for performing a range of transformations on input samples. However, unlike AutoML-based methods, Smart Augment does not include an optimization and selection stages to find effective augmentations.

5.2.2. Optimization techniques for meta-learning augmentations

An important aspect of the automated augmentation problem is determining the best possible combination of the different augmentation operations that result in optimum performance. Solving this sub-task is challenging to carry out manually because of the enormous computational requirements. In autoML-based data augmentation, this problem is formulated as an optimization problem and an appropriate search algorithm is then used to find the best solution. The goal of the optimization algorithm is to find the policy that yields the best performance on the validation data. In general, the algorithms for solving this ill-formalized optimization problems are not required to converge to a strictly optimal solution. Indeed, most AutoML methods are design to find acceptable augmentation policies in a reasonable time frame. We describe the main techniques and the specific optimization approaches commonly used to accomplish data augmentation.

Reinforcement learning: Many auto-ML-based meta-learning approaches (e.g. [85,253,259–261]) – including AutoAugment [250], the original work proposed by Cubuk et al.– employ Reinforcement learning (RL) techniques to optimize augmentations. With RL approach, the model is rewarded for applying augmentation policies that yield high accuracy and punished for selecting policies that result in low accuracy. This encourages the model to learn to apply better policies over time.

Bayesian optimization: Recently, approaches (e.g., [252,262]) based on Bayesian optimization techniques have shown better generalization performance with less computational overheads. Bayesian optimization is a sequential probabilistic modeling technique that constructs a probabilistic model to predict the best hyperparameters for a given machine learning task. The predicted hyperparameters are then used to evaluate performance based on an objective function. The Bayesian learning process involves tracking previous evaluation results and incorporating new information to refine the probabilistic model at each iteration.

Evolutionary algorithms: As an alternative to the Reinforcement learning and Bayesian approaches, many works [263–266] have proposed evolutionary computational techniques [267] for automating data augmentation. One of the first works to use this method for data augmentation was [265]. Evolutionary computation methods search within the set of augmentation policies for several possible solutions rather than for a single best policy. At each iteration of the process, bad policies are eliminated while good ones are retained and refined further. This procedure ultimately finds the best augmentation policies from the initial set of primitive transformation operations.

Gradient-based optimization of augmentation parameters: More recently, a number of works [71,253,254,268] have proposed to use gradient-based optimization algorithms that utilize backpropagation to find the best augmentations. Because the search space (i.e., the set of primitive transformation operations) are inherently discrete, the use of gradient-based techniques is not straightforward. In most cases, it is often necessary to modify the representation of the search space so that numerical approximations could be used to estimate gradient for the optimization process.

6. Effectiveness of data augmentation

In this section, we discuss the effectiveness of different augmentation strategies. It must be noted that many state-of-the-art deep learning models generally employ a combination of data augmentation methods to extend training data. However, these works do not usually disentangle the impact of the constituent augmentation operations. Because of this, in most works it is generally challenging to determine the effectiveness of individual augmentation methods on performance. Moreover, in many cases, they impact of data augmentation is ignored altogether since the focus is usually on the algorithmic techniques under study. Nonetheless, a number of studies have reported performance on specific data augmentations methods. Many works have also investigated the impact of combining multiple augmentation operations to expand training data. We analyze the results of these works to provide insights into the comparative strengths and weaknesses of various augmentation strategies in different situations.

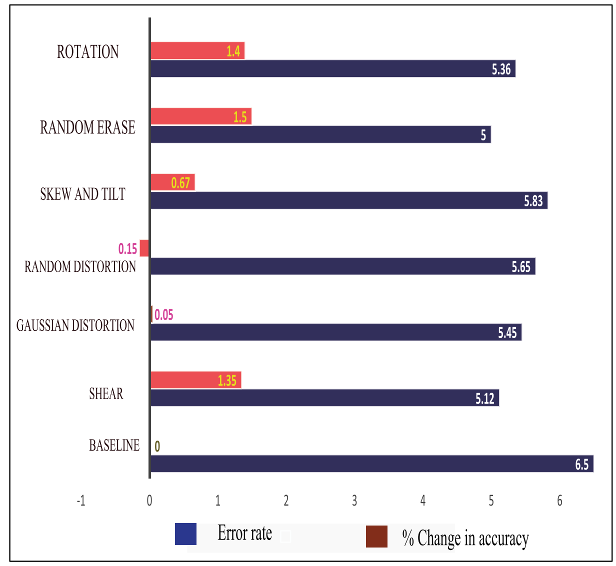

Fig. 19. Pedictive performance results for single augmentations on CIFAR-10 using ResNet-56 model.

For clarity of presentation, we categorize all data augmentation methods into four broad groups — (1) classical approaches that employ a single generic transformation operation to achieve augmentation, (2) non-traditional augmentation approaches that employ advanced techniques to transform data, (3) methods that rely on synthetic data to extend training samples, and (4) augmentation approaches that make use of different combinations of the methods in the first three groups.

To enable fair comparison of different approaches, we focus on experiments that have been conducted under similar conditions — i.e., same backbone networks and datasets, similar training methods and times (in terms of number of epochs). We summarize the results of several studies in Tables 1 and 2 for comparison. In the cases of generic data augmentation, data synthesis and combined augmentation techniques, standard settings have not been broadly used to test the different methods in a way that provides a fair comparison of the methods. Because of this limitation, for these approaches, we discuss individual studies that investigate the relative performance of different augmentation methods.

6.1. Effectiveness of generic (basic geometric and photometric) data augmentation strategies

One of the most comprehensive performance evaluations of basic data augmentation strategies has been carried out in [276]. In the work, O'Gara and McGuinness conducted comparative studies on the performance of seven different geometric transformations on the CIFAR-10 using ResNet-56 model as the baseline. These were: rotation with strengths between −25° to +25°; skewing and tilting in different directions between 0 and 22.5°; shearing along the x-axis or y-axis with −20 to +20°; random distortions, whereby distortions specified by grid width and height and distortion magnitude are randomly applied to different regions of some images; Gaussian distortion that applied distortions using 2D normal distribution to the rectangular grid patches; and random erasing — where RCB components of rectangular image regions (up to 50% of image pixels) are set to random values. The results of their study are summarized in Fig. 19.

From the Figure 19, it can be observed that random erasing augmentation approach provides the best performance improvement (1.5

Table 1
Performance results of various advanced data augmentation methods on CIFAR-10 and CIFAR-100. The methods have been tested using three CNN models — WideResNET28-10 (WRN), PyramidNet (PNet) and Shake-Shake 26-32 (S-S).

Method	CIFAR-10				CIFAR-100			
	WRN	S-S	PNet	Mean	WRN	S-S	PNet	Mean
Baseline	96.1	97.1	97.3	96.83	81.2	82.9	86.0	83.37
MixUp [96]	–	–	–	–	82.1	84.8	–	83.45
CutMix [92]	–	–	–	–	82.9	85.0	–	83.95
SuperMix [109]	–	–	–	–	83.6	85.5	–	84.55
SuperMix+ [109]	–	–	–	–	83.9	85.8	–	84.85
AA [250]	97.4	98.1	98.5	98.0	82.9	85.7	89.3	85.97
FastAA [252]	97.3	98.0	98.3	97.87	82.8	85.4	88.3	85.50
FasterAA [253]	97.4	98.0	–	97.70	82.2	84.4	–	83.30
RA [251]	97.3	98.0	98.5	98.0	83.3	–	–	83.30
RE [269]	96.9	–	–	96.9	82.3	–	–	82.30
HaS [80]	96.9	96.9	–	96.9	–	–	–	–
ISDA [134]	97.5	–	–	97.5	82.0	–	–	82.0
PBA [265]	97.4	98.0	98.5	97.97	83.3	84.7	89.1	85.70
AdvAA [259]	98.1	98.2	98.6	98.3	84.5	85.9	89.6	86.67
IHDA [143]	97.8	98.3	98.8	98.3	86.1	88.8	90.3	88.30
IHDA+ [143]	97.9	98.3	98.8	98.3	86.8	88.9	91.0	88.90
RUA [62]	97.4	–	98.5	97.4	83.6	–	–	83.6
DADA [268]	97.3	98.0	98.3	97.9	82.5	84.7	88.8	85.33
DeepAA [270]	97.4	98.1	–	97.8	83.7	85.2	–	84.45
MetaAugment [271]	97.7	98.3	98.6	98.2	83.8	86.0	89.5	86.43
TA [272]	97.5	98.2	98.6	98.1	84.3	86.2	–	85.25
PAA [85]	97.8	98.2	–	98.0	83.3	–	–	83.3
DivAug [273]	98.1	98.1	98.5	98.2	84.2	85.3	–	84.75
OLHA [261]	97.4	–	–	97.4	–	–	–	–
AWS [260]	98.0	98.3	98.7	98.3	84.7	85.9	89.6	86.73
Cutout [77]	96.9	97.4	97.7	97.3	81.6	84.0	87.8	84.47
UA [274]	97.3	98.1	85.5	93.6	82.8	85.0	–	83.90
DDAS [254]	97.3	97.9	–	97.6	83.4	84.9	–	84.15
GridMask [83]	97.2	97.2	–	97.2	–	–	–	–
RICAP [106]	97.1	97.9	–	97.4	–	–	–	–
SaliencyMix [105]	96.0	–	–	96.0	80.5	–	–	80.5

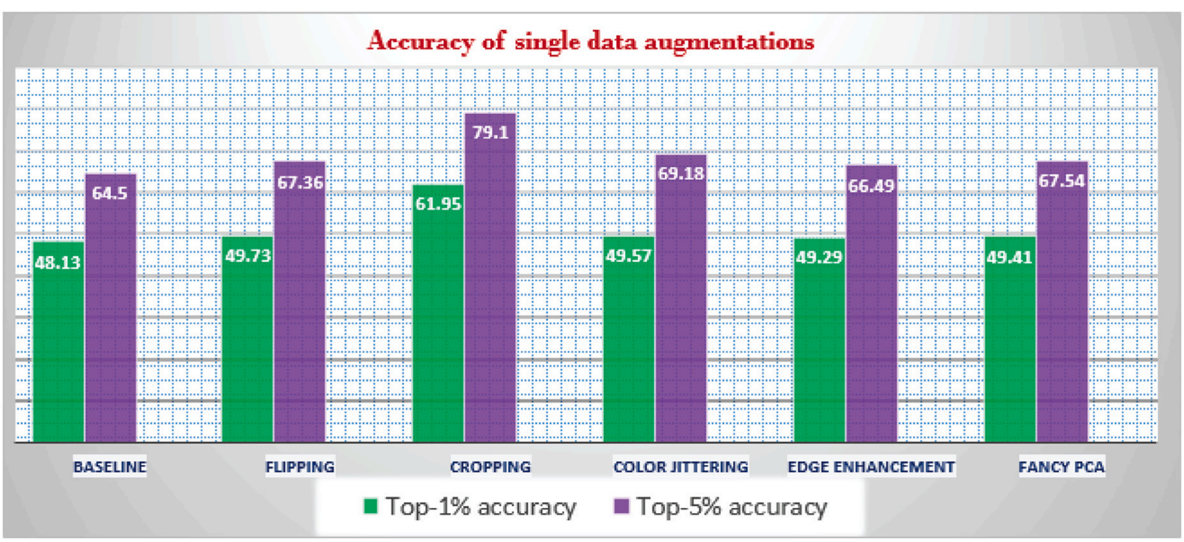

Fig. 20. Results of comparison of different traditional data augmentation methods on Caltech 101 dataset, conducted by [275].

over the baseline), obtaining between 0.12 and 1.65% better generalization accuracy than the other methods. Shearing (1.35%) and rotation (1.14%) have also shown appreciable performance gains. On the other hand, distortion augmentations is the least effective method, yielding only marginal or no improvement at all. The poor performance of image distortion augmentations can be explained by the fact that the target tasks in this domain do not usually contain such distorted images. This shows that including irrelevant augmentations not only fails to achieve improvements, but can indeed harm performance. The authors in [276] also showed that the timing as to when to introduce augmentations in the training process is also important to the overall performance gains achievable. The total number of epochs for the entire training was 163. Interestingly, all seven augmentations recorded their best performance when augmentations are introduced during the 30th epoch of training. This phenomenon needs further investigation as its cause has not been established.

Taylor and Nitschke [275] (see Fig. 20) also obtained higher performance improvements when using geometric transformations on the Caltech 101 dataset (between 1.6 and 13.8 increase in top-1 accuracy) compared to augmentation strategies based on photometric transformations (1.2–1.4 increase in top-1 accuracy). In their work, cropping resulted in the best overall accuracy.

Table 2
Results of advanced data augmentation strategies on ImageNet dataset. The result are based on tests using ResNet-50 and ResNet-200 models. Going by common practice, for ImageNet, we present results for top-5 and top-1% accuracies.

Method	ResNet-50				ResNet-200			
			Acc. improvement				Acc. improvement	
	Top-1	Top-5	Top-1	Top-5	Top-1	Top-5	Top-1	Top-5
Baseline	76.3	93.1	–	–	78.5	94.2	–	–
MixUp [96]	77.0	93.4	+0.7	+0.3	79.6	94.8	+1.1	+0.6
CutMix [92]	77.2	93.5	+0.9	+0.4	79.9	94.9	+1.4	+0.7
SuperMix [109]	77.6	93.7	+1.3	+0.6	80.8	95.4	+2.3	+1.2
SuperMix+ [109]	78.2	94.0	+1.9	+0.9	81.3	95.6	+2.8	+1.4
AA [250]	77.6	93.8	+1.3	+0.7	80.0	95.0	+1.5	+0.8
FastAA [252]	77.6	93.7	+1.3	+0.6	80.6	95.3	+2.1	+1.1
RA [251]	77.6	93.8	+1.3	+0.7	–	–	–	–
HaS [80]	77.2	–	+0.9	–	–	–	–	–
PBA [265]	77.2	93.4	+0.9	+0.3	–	–	–	–
AdvAA [259]	79.9	94.5	+3.6	+1.4	81.3	95.3	+2.8	+1.1
IHDA [143]	79.2	95.1	+2.9	+2.0	82.0	96.8	+3.5	+2.4
IHDA+ [143]	79.9	95.9	+3.3	+2.8	82.9	97.3	+4.4	+3.1
AdvAA	79.4	94.5	+3.1	+1.4	81.3	95.3	+2.8	+1.1
RUA [62]	77.7	–	+1.4	–	–	–	–	–
DADA [268]	77.5	93.5	+1.2	+0.4	–	–	–	–
MetaAugment [271]	79.7	94.6	+3.4	+1.5	81.4	95.5	+2.9	+1.3
TA [272]	78.1	93.9	+1.8	+0.8	–	–	–	–
PAA [85]	77.5	–	+1.2	–	–	–	–	–
Puzzle Mix [104]	77.5	93.8	+1.2	+0.7	–	–	–	
OLHA [261]	78.9	94.3	+2.6	+1.1	–	–	–	–
AWS [260]	79.4	94.5	+3.1	+1.4	81.4	95.3	+2.9	+1.1
StyleMix [237]	75.9	92.9	−0.4	−0.2		–	–	
DropBlock	78.3	94.1	+2.0	+1.0	–	–	–	–
Manifold Mixup [147]	77.5	93.8	+1.2	+0.7	–	–	–	–
mWh [113]	76.9	93.5	+0.6	+0.4	–	–	–	–
DropPath [277]	77.1	93.5	+0.8	+0.4	–	–	–	–
AugMix [116]	76.7	93.3	+0.4	0.0	–	–	–	–
SmoothMix [112]	77.7	93.6	+1.4	+0.5	–	–	–	–
StochasticDepth [156]	77.5	93.7	+1.2	+0.6	–	–	–	–
SaliencyMix [105]	78.7	–	+1.4	–	–	–	–	–

Mash et al. in [278] evaluated a wide range of geometric data augmentation techniques on image classification tasks. They used Af-Caffe model (a derivative of AlexNet [279] developed in [280]) and a training dataset that consisted of 76,426 aircraft images taken from various online sources such as Youtube and Flickr. The specific methods evaluated were horizontal flipping, rotation, polygonal occlusion (i.e., masking random image regions with polygons of specified number of vertices), cropping and scaling. In addition, different combinations of the aforementioned methods were tested. Some of their methods provided a gain of as much as 45% improvement over the augmentation strategies used in [279]. Also, the authors reported that randomly cropping fixed sized patches as implemented in [279] provides minimal benefits compared with random and variable size cropping. Experimental results show that, overall, the polygon occlusion strategy with 9 vertices applied in conjunction with random cropping resulted in the highest performance gain (9.1%), followed by variable size random cropping (6.1%), variable size random cropping with horizontal flipping (5.6%), 9 vertex polygon occlusion (4.9%), and so forth. From the results, the performance of the polygon occlusion method was also noticeable. The effectiveness of this particular type of augmentation operation in this scenario, most likely, stems from the fact that many of the test images contain cases of aircraft refueling scenes, where refueling booms often occlude aircrafts with a geometrically similar random patterns. The authors hypothesized that this type of random occlusions may provide a dropout-like effect on the network but this ignores the fact that such an effect could still be achieved with standard occlusion patterns. Again, the superior performance of geometric transformations here is consistent with the results of Taylor and Nitschke [275], O'Gara and McGuinness [276], and Qin et al. [281].

For tasks such as detection, some works [282,283] propose to extend the training set used to train the backbone classification models instead of directly augmenting the dataset for object detector. Chen et al. in [282] investigate the efficacy of this approach extensively.

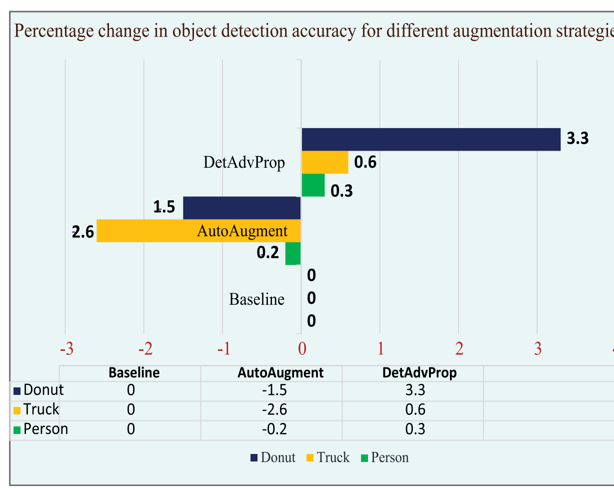

Fig. 21. Carrying out augmentation on the pre-trained classification backbone has been shown to improve performance. However, in some tasks, this can potentially reduce detection accuracy on the end task. Proposed in [282] appears to alleviate this problem.

While the approach improves performance in the end tasks in some cases (e.g., [284]), their results show the gains from augmenting the backbone models do not readily transfer to the end detection task after making specific fine-tuning for object detection. In particular, augmenting dataset for classical backbone models using AutoAugment has been shown to harm performance on the end-task. To address this shortcoming, the authors in [282] propose a new method, Det-AdvProp, that employs adversarial training to provide model-agnostic augmentation in the fine-tuning phase and achieve a significant performance

Table 3
Effectiveness of traditional and image synthesis approaches to data augmentation. Results are from [286].

Task	Model	Augmentation approach			
		None	Trad.	Style	Both
AW→ D	InceptionV3	0.789	0.890	0.882	0.952
	ResNet18	0.399	0.704	0.495	0.873
	ResNet50	0.488	0.778	0.614	0.922
	VGG16	0.588	0.830	0.551	0.870
	Average	**0.551**	**o.801**	**0.636**	**0.904**
DW → A	InceptionV3	0.183	0.160	0.254	0.286
	ResNet18	0.113	0,128	0.147	0.229
	ResNet50	0.130	0.156	0.170	0.244
	VGG16	0.086	0.149	0.111	0.243
	Average	**0.128**	**0.148**	**0.170**	**0.251**
AD → W	InceptionV3	0.695	0.733	0.767	0.884
	ResNet18	0.414	0.600	0.424	0.762
	ResNet18	0.491	0.676	0.508	0.825
	VGG16	0.465	0.679	0.426	0.752
	Average	**0.516**	**0.672**	**0.531**	**0.806**

boost in object detection tasks. The results of the study are summarized in Fig. 21.

6.2. Effectiveness of advanced data augmentation methods

In this section, we summarize performance results for some of the advanced data augmentation methods discussed in Sections 2, 3, 4 and 5. For a fair comparison of the different methods, we focus on approaches that use the same dataset, model configuration and training settings. In Table 1, we present quantitative performance results for several data augmentation methods on CIFAR-10 and CIFAR-100 datasets. Results on ImageNet dataset are presented in Table 2. Going by common practice, for ImageNet, we present results for top-1% and top-5% accuracies. For ease of comparing the data, we represent the mean gain in performance (i.e., percentage change in accuracy relative to the baseline averaged over the different models tested for the particular dataset). The results show an appreciable performance improvement over the baseline configurations for almost all the methods. The baselines employ off-the-shelf deep CNN models with only minimal augmentations based on basic geometric and photometric transformations. The methods on CIFAR-10 and CIFAR-100 have been tested using three different state-of-the-art deep networks: Wide ResNet 28–10 (WRN28-10) [153], Shake-Shake 26-32 [128] and PyramidNet [154]. Results on ImageNet were all obtained using ResNet-50 and ResNet-200 models [285].

In the tables, some works (indicated by "+" sign) simultaneously present results of advanced methods in addition to their implementation with additional basic augmentations based on traditional approaches. For example, Khan and Fraz [143] evaluated IHDA and another version trained with extra augmentations (i.e., IHDA+). The results show good performance gains can be obtained when traditional methods are used in combination with the new approaches. However, the gains are not consistent across all datasets. While ImageNet and CIFAR-100 provided significant improvements, overall performance on CIFAR-10 was only marginally better than the [143] IHDA version without extra augmentations.

6.3. Effectiveness of combined augmentations

In many practical tasks, not one but many different data augmentation operations are typically applied simultaneously. This can lead to significant performance improvements even on very large-scale datasets. For datasets that lack the sufficient variability, additional data augmentation strategies often lead to significant performance improvements. For example, Paulin et al. [287] achieved up to 4.8% improvement in image classification accuracy on ImageNet dataset after using a combination of data augmentation strategies. On CUB, a relatively small dataset, the gain was 17.0%. We look at the effect of employing multiple augmentations in this sub-section. Shijie et al. [288] compared the effectiveness of three broad classes of augmentation methods: GAN, geometric transformations (specifically, flipping, rotation, shifting and cropping) and photometric methods (color jittering, PCA and noise injection). They also tested different permutations of the individual augmentations in pairs. In addition, they assessed the influence of different proportions of augmented data on performance of deep neural networks that utilized these augmentations. Among the individual augmentations, the best performing augmentations were found to be WGAN, flipping, cropping and rotation. These methods were subsequently combined and tested in pairs, with Flipping/WGAN pair emerging as the best performer. Indeed, all the paired augmentations outperformed the individual ones. With the exception of the case of noise injection, training with a larger set of augmented samples consistently performed better. For instance, in the study, training with augmented images three times the number of clean ones achieves a higher accuracy than with twice the number of clean sample, which also did better than training with equal number of clean and augmented images. Moreover, the impact of data augmentation was more pronounced when the original training dataset was small.

Jackson1 et al. [286] compared performance gains from style transfer method and a combination of several different traditional augmentation strategies on different machine vision tasks. The suit of traditional augmentation strategies involved both photometric and geometric image transformations. The photometric methods were brightness and contrast changing, color jittering (HSV space), and color space conversion (HSV to grayscale). The geometric augmentations used included shearing, cropping, rotation, and horizontal flipping. In addition, random erasing, which we categorized in this survey as region elimination technique, was also employed as part of the traditional approaches. They tested these data augmentation methods on cross-domain classification tasks, were the Office dataset [289] was used. The dataset is a cross-domain (specifically, a three-domain) dataset consisting of pictures of Amazon products listings, as well as Webcam and DSLR images. The entire dataset comprises 4,110 images grouped in 31 categories of common office objects. To test cross-domain generalization accuracy, two of the dataset domains are combined for training and transferred to the other during inference. The results obtained by different deep CNN models trained with the data augmentation methods are shown in Table 3. These results show that even for cross-domain tasks, traditional data augmentation techniques are very powerful. In fact, they are even significantly more powerful than the more advanced style transfer method when each is used separately. A drastic performance gain is achieved when the two classes of approaches are used together. The average gain in generalization accuracy is 35.3% when trained on AW and perform inference on D ($AW \rightarrow D$); 12.3% when trained on DW and perform inference on A ($DW \rightarrow A$), and 29.0% when trained on AD and perform inference on W ($AD \rightarrow W$). By contrast, the average gains from using traditional and style augmentations separately are significantly lower — 25.0 and 8.5% for $AW \rightarrow D$; 2.0 and 4.2% for $DW \rightarrow A$; and 15.6 and 1.5% for $AD \rightarrow W$. The impressive gain in accuracy is almost additive. It can be hypothesized that the role of style augmentation is to compensate for texture dependence [235], and since this attribute is complementary to the effect of the traditional data augmentation methods, the gains resulting from the two sets of approaches invariably add up, leading to significant performance gain.

Ornek and Ceylan in [290] investigated the effectiveness of different traditional data augmentation methods on classification accuracy of medical thermography images — that is, images captured using thermal cameras. The data used contain only real-world thermal images collected from a medical facility. The authors compared the application of ten different augmentations – four geometric transformations (rotation, zooming, mirroring and shearing) and six photometric transformations

Table 4
Quantitative results on the performance of different combinations of geometric and photometric data augmentation methods (obtained by Ornek and Ceylan in [290])

Exp.	Data augmentation method			Acc.
	Method 1	Method 2	Method 3	
1	None	None	None	73.55
2	Rotation	Mirroring	zooming	64.26
3	Rotation	shearing	zooming	70.88
4	shearing	blurring	Mirroring	78.76
5	Hist. Equal.	zooming	brightness	85.36
6	Rotation	blurring	brightness	85.86
7	Sharpening	shearing	Color adjust.	86.96
8	zooming	Color adjust.	brightness	92.03
9	Contrast	brightness	Sharpening	93.40
10	Hist. Equal.	Color adjust.	Rotation	94.46
11	Contrast	Sharpening	blurring	99.84

Table 5
Performance of traditional data augmentation strategies and neural style transfer methods on Caltech 101 dataset using VGG16 (as obtained by Zheng et al. [226].

Approach	Augmentation	Acc.
Traditional	None (baseline)	0.8334
	Flipping	0.8305
	Flipping + Rotation	0.77
Style Transfer	Snow	0.8526
	RainPrincess	0.8494
	Scream	0.8490
	Wave	0.8468
	Sunflower	0.8461
	LAMuse	0.8436
	Udnie	0.8404

(histogram equalization, color jittering, blurring, sharpening, brightness adjustment and contrast changing) – in groups of three combinations as shown in Table 4. In the study, it was observed that the application of three geometric transformations in combination negatively impacts generalization performance. In particular, applying only rotation, mirroring and zooming results in a significant performance degradation from 73.55% accuracy on the baseline (unaugmented) dataset to a mere 64.26% when these three geometric transformations were used. Indeed, in all cases where geometric transformations were employed, even in cases where only one geometric transformation was used in conjunction with two photometric transformations, performance was markedly inferior to all the strategies that relied on only photometric augmentations. For instance, applying a combination of sharpening, blurring and contrast adjustment augmentations improves performance by 26.29% (from 73.55 baseline to 99.84%), whereas the maximum attainable result when any geometric augmentation is used is 94.46%. This clearly shows that photometric transformations were more useful than geometric transformations in the context of thermal image recognition in real-world medical settings. The use of geometric transformations seems to harm performance in this particular application scenario. This poor performance of geometric transformations may be attributed to the controlled settings in which the images are captured and used (i.e., when the inference system is presented with images of the same, pre-determined orientation). Experiments on the efficacy of geometric augmentations for COVID-19 detection, conducted by Elgendi et al. in [291], also proved that geometric transformations produce poor results in controlled settings. In the study, all five geometric augmentation strategies tested – flipping, shearing, rotation, scaling and translation – appeared to harm performance. Indeed, the worst performance is obtained when all five geometric augmentations are applied simultaneously.

In contrast to the experimental results of [290], the classification performance obtained by Taylor and Nitschke [275] on Caltech 101 dataset (Fig. 20) showed that geometric transformations were more effective than photometric transformations. This outcome may stem from the fact that the Caltech 101 contains images taken from arbitrary view angles, where spatial variations can be pronounced. As a result, data augmentation strategies employing spatial warpings can provide invariance to these effects. In one of the most comprehensive studies [292], Kim et al. investigate several data augmentation techniques based on a special form of color jittering method called (PSNR) described in Section 2.1.2. The PSNR augmentation technique was applied on three different color spaces (RGB, HSV and CIE L*a*b*). The augmented data consisted of images of these three color spaces — named PSNR RGB, PSNR HSB and PSNR Lab. Further, two different geometric transformations, rotations and flipping, were applied. The authors carried out extensive experiments to determine the best augmentations and their most effective combination suitable for visual inspection in manufacturing settings, where the number of training images is often very small. Like in [290], they tested the performance of potentially useful augmentation strategies by considering each augmentation separately and then experimenting with different permutations of the individual augmentation strategies. They first demonstrate that PSNR-based color space data augmentation method outperforms conventional color jittering methods by up to 39% (63% vs 44%) when applied to small training samples of less than 3,000 images. Further, the authors in [292] investigated the effect of applying geometric transformations (flipping and rotation) on the datasets augmented by color space perturbation. The experimental results show that rotations provide higher performance improvements than flipping. This can be attributed to the nature of the task — visual recognition of objects in a manufacturing process, where changes in object orientations are more likely to be encountered. Again, this shows that task-relevant augmentations may be key to performance improvements in many application settings. Applying the two augmentations together with the color space transformation results in only a moderate increase in generalization performance. This result also underscores the limit of stacking several data augmentation together. Indeed, adding useless augmentations can actually harm performance.

In another classic work [226] Zheng et al. compared the impact of two different geometric transformations (rotation and flipping) and a more advanced data augmentation method based on neural style transfer [223]. Their experiments were conducted on Caltech 101 and 236 datasets with VGG backbone. They obtained results for single and multiple augmentations. The style transfer methods were seen to be more useful strategies. It was observed that the two geometric transformations penalize performance significantly when applied together (Tables 5 and 6), recording a performance degradation of 6.34% (from 83.34% in the unaugmented training set to 77.00% accuracy in the set with both flipping and rotation).

It is evident from Table 6 that style transfer augmentations improves performance in all instances in which they are combined with traditional geometric augmentations. One limitation of the study, however, is that the authors did not investigate the effect of combining different styles without geometric transformation applied. One can, however, infer from the results that using multiple styles in combination would increase generalization performance. Indeed, in one instance in [226] (Table 6) both flipping and wave style reduced performance when applied separately but their combination resulted in a significant performance gain.

7. Summary and discussion

Data augmentation remains one of the most important tasks in the machine learning development process. In general, data augmentation is accomplished by either transforming training samples in some desired manner for use as additional training data or by creating completely new samples from scratch. Transformation operations can be applied directly on input data or indirectly by manipulating feature vectors in intermediate layers of deep neural networks. The common

Main approach	Description	Common techniques	General strengths	Weaknesses
Input space DA	Transformation of input images with analytical or learned operations	• Geometric transformation • Photometric transformation • Region and pixel level augmentation operations	• Easy to implement • More predictable action • More intuitive	• Relatively expensive • Can lead to increased memory demands
Feature space DA	Manipulation of deep feature vectors to enhance variability	• Feature mixing • Feature dropping • Feature interpolation	• More computationally efficient • Can produce more granular visual attributes	• Impact less predictable beforehand • Can negatively affect adversarial robustness
Data synthesis	Generation of new data from scratch	• Computer graphics modeling • Neural rendering • Generative modeling • Neural style transfer	• The only viable solution in cases with no data • Mostly more flexible and scalable	• Laborious • Domain gap between real and synthetic data • Subject to limitations of developers' biases & imagination • Lack of quality metrics
Meta-learning	Learning generalized representation of data Learning policies to find effective augmentations automatically	• Reinforcement learning • Genetic algorithms • Bayesian optimization • Random and greedy search	• Less dependent on intuition • Can avoid limitations associated with bias	• Computationally expensive • Currently limited to only transformation-based DA

Fig. 22. A broad summary of the main categories of methods discussed in this work.

Table 6
Performance of traditional and neural style transfer methods on Caltech 101 and 236 datasets using VGG19. Results are from experiments conducted in [226].

Traditional method	Style transfer	Acc.
Results with VGG19 and Caltech 101		
None	None	0.8450
Flipping	None	0.8367
None	Wave	0.8425
Flipping	Wave	0,8581
None	Scream	0.8446
Flipping	Scream	0.8465
Results with VGG19 and Caltech 236		
None	None	0.6200
Flipping	None	0.6666
None	Scream	0.6313
Flipping	Scream	0.6728
None	Wave	0.6272
Flipping	Wave	0.6632
Combined geometric (VGG19 and Caltech 101)		
Flipping Rotation	None	0.7521
Flipping Rotation	Wave	0.7837
Flipping Rotation	Scream	0.7862
Flipping Rotation	Scream Wave	0.7942

types of input space data augmentation methods use basic image transformation operations to perform geometric transformations such as rotation, flipping, cropping and scaling. In addition, photometric transformations like blurring, color jittering, noise injection and removal, color space conversion, as well as brightness, contrast and other pixel content adjustment techniques like fancy PCA are also common. In most practical applications, generic photometric and geometric transformations are often used as the first line of measures to enrich training data. Together, these two types of augmentation methods provide complementary benefits in many general computer vision settings. Recently, non-traditional image processing-based data augmentation methods such as random erasing, image mixing and swapping have shown a lot of promise. Some new approaches rely on learned transformation operations to apply appropriate augmentations. Even though these are powerful, they are generally more computationally demanding and laborious to construct. In most cases, simple and standard augmentation schemes give competitive results and can be effectively used as a viable alternative to the more complex ones. It is common to first apply more basic augmentations before employing the advanced methods to further push the performance ceiling beyond what can reasonably be achieved by these basic techniques.

In contrast to input space augmentation approaches that produce augmentations by transforming input samples explicitly (i.e., using analytical formulations) or implicitly (i.e., using learned transformation operations) to create new variations, feature space data augmentation strategies rely on heuristics and learned mechanisms to transform and aggregate output features in various ways in order to enhance variability. A key advantage of deep feature level augmentation lies in the fact that the dimensionality and complexity are lower than at the raw input level and can, thus, be manipulated more easily to obtain useful and diverse visual properties of the underlying input images. Moreover, deep features are able to better capture subtle factors of image variations than can be achieved with input level augmentation methods. Notwithstanding these advantages, data augmentation approaches that transform training data at the input (i.e., image) level are more intuitive and are preferred in application domains where known image transformations are expected.

Data augmentation techniques based on synthetic data generation are useful for training machine learning models in settings where there is no data or where data acquisition cost is exorbitant. Key requirements of image synthesis methods are the ability to generate natural visual attributes and their correct distribution in the target environments. In addition, the data must be easily synthesizable so that large quantities of training samples can be generated to meet the application needs. Synthetic data generated by 3D modeling tools and differential neural rendering techniques provide more scalable resolutions for different needs. They also readily support more explicit and precise labeling of the generated data. In last few years, generative modeling methods, especially techniques based on generative adversarial networks, have become the most predominant means of generating synthetic data for training deep learning models. These techniques leverage easily accessible unlabeled data to train models to generate new data. The use of generative modeling methods is, however, severely restricted in situations where large corpus of training data cannot be obtained. In such situations, generative modeling methods may not be practical or would themselves require data augmentation to be realized. A major shortcoming of all data synthesis approaches is the lack of reliable evaluation metrics to compare the quality of artificial data to real ones with respect to a given task. The main approaches, their essence, as well as the common strengths and weaknesses are summarized in Fig. 22.

8. Key challenges and promising research directions

Based on the extensive survey of state-of-the-art data augmentation strategies, we summarize some of the main open problems and the most promising directions for future research.

- *Towards physically grounded augmentations and semantically meaningful control of visual attributes:* Data augmentation for machine vision tasks is commonly achieved by manipulating images in 2D space. In many real-world applications, however, data augmentation techniques based on this approach have serious limitations because of their inability to simulate the complex visual properties of the 3D world. Recent advances in approaches based on implicit neural representation, differential neural rendering, neural radiance fields (NeRFs) and 3D-aware generative modeling makes it possible to introduce realistic behavior that reflects real-world properties of objects and scenes such as geometry, motion, lighting, color, shadows, reflections, texture, pose and weather. In the future, these approaches would increasingly enabled more powerful representations and provide high level control over the visual properties of artificially generated data. This enhanced representation power of deep learning models and additional control can be leveraged to specifically generate desired visual appearance based on context-relevant semantic information. In autonomous driving, for example, such a capability would allow visual properties of realistic, large-scale dynamic scenes to be adapted on the fly according to prevailing environmental conditions such as varying weather, time of the day, or seasons.
- *Determining useful augmentations and their effective combinations without costly experimentations:* In most computer vision applications, multiple augmentations are usually required to achieve the required level of performance. However, as evidence from many studies discussed in the survey shows, certain combinations of augmentations are usually harmful to overall performance. Presently, determining beforehand the set of augmentations that would yield optimal performance in a given situation is still an open problem. We expect advances to be made in this direction to provide more theoretical background to the selection of effective augmentation for specific tasks. This would obviate the costly and time-consuming process of trial and error means of determining useful augmentation strategies, which can often lead to sub-optimal strategies.
- *Extending the scope of meta-learning-based data augmentation techniques:* Automated machine learning methods have shown a lot of promise, but so far they have generally been limited to data transformation-based augmentation methods. Only a few works have attempted to apply AutoML techniques to data synthesis. Even in this domain they are mostly limited to finding effective augmentations by optimizing manually constructed image transformation operations. Future research will increasingly advance algorithms for automating the very process of generating the primitive operations required to perform data synthesis. It is expected that in the near future, as these approaches continue to evolve, their scope of application within the context of data augmentation will expand. It is also expected that future work will focus on extending AutoML techniques to many other data-related sub-tasks of machine learning, including data acquisition, preparation and pre-processing.
- *Towards better quality metrics for the evaluation of training data:* One of the biggest challenges of machine learning today is the lack of suitable metrics for evaluating the quality of training data. A means of determining the quality of data is particularly useful in scenarios where the data is not obtained from the natural environment or by natural processes but is artificially generated. It is expected that metrics that allow to objectively determine the quality of a particular dataset, especially synthetic ones, for a target environment would become practical in the near future. This would help model developers and researchers to refine data at the synthesis stage rather than conducting elaborate evaluations through training and testing.

9. Conclusion

Data is crucial for all deep learning-based systems, regardless of the domain or application, and its quality ultimately determines the overall performance and reliability of the underlying models. Data augmentation is a way to extend training data in order to allow machine learning models to achieve good results in situations where training data is limited or is of poor quality or even absent. In this paper, we present a comprehensive survey of data augmentation methods in computer vision. The survey overs general principles, use-cases and various application issues of modern data augmentation techniques. The main approaches surveyed include traditional approaches as well as modern techniques based on region-level transformation and feature space augmentation, as well as data synthesis methods such as CAD modeling, neural style transfer, generative modeling and neural rendering. Meta-learning approaches based on traditional concepts and automated machine learning (AutoML) are also covered. In addition, we discuss the effectiveness of different data augmentation approaches to guide developers and researchers in choosing appropriate augmentation methods for different tasks. Although data augmentation methods based on explicit transformation operations provide accurate and reliable performance improvements, in many computer vision tasks, serious difficulties arise in their practical implementation. In particular, the challenge of knowing beforehand which transformations would work best and the fact that these approaches still require training examples are key among these difficulties. Data synthesis approaches alleviate the last problem but they still suffer from difficulty of predicting performance. In order to know which transformations or combinations of transformations work best, it is often necessary to experiment with various strategies using some of the real data for validation. This leads to a time-consuming process of trial and error. Although automated data augmentation techniques have been proposed to solve this problem, currently, the range of augmentation operations supported by these methods is limited. In addition, they introduce increased computational costs and additional memory requirements. These costs impose restrictions on the volume of data that can be generated by these advanced augmentation techniques. In many situations, from the computational cost perspective, automated data augmentation may not be justified. Consequently, research on effective data augmentation methods is still very active.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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